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(12) **United States Patent**
Schomburg et al.(10) **Patent No.: US 12,409,219 B2**(45) **Date of Patent: Sep. 9, 2025**(54) **RECOMBINANT POLYPEPTIDES
CONTAINING AT LEAST ONE
IMMUNOGENIC FRAGMENT AND USES
THEREOF**(71) Applicant: **Boost Biopharma, Inc.**, Woburn, MA
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(US)(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 721 days.(21) Appl. No.: **17/728,798**(22) Filed: **Apr. 25, 2022**(65) **Prior Publication Data**

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Related U.S. Application Data(63) Continuation-in-part of application No. 17/535,309,
filed on Nov. 24, 2021, now Pat. No. 11,358,990,
which is a continuation of application No.
PCT/US2021/040019, filed on Jun. 30, 2021.(60) Provisional application No. 63/154,647, filed on Feb.
26, 2021, provisional application No. 63/046,426,
filed on Jun. 30, 2020.(51) **Int. Cl.****C07K 14/005** (2006.01)**A61K 39/00** (2006.01)**A61K 39/215** (2006.01)**A61K 47/02** (2006.01)**A61P 31/14** (2006.01)**C07K 14/165** (2006.01)**C07K 16/10** (2006.01)(52) **U.S. Cl.**CPC **A61K 39/215** (2013.01); **A61K 47/02**
(2013.01); **A61P 31/14** (2018.01); **C07K**
14/165 (2013.01); **A61K 2039/55511** (2013.01)(58) **Field of Classification Search**CPC **C07K 14/005**; **C07K 2319/30**;
C07K 14/165; **C07K 2319/00**; **C12N**
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See application file for complete search history.

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Ltd.(57) **ABSTRACT**Provided is a recombinant polypeptide containing at least
one immunogenic fragment of Severe Acute Respiratory
Syndrome Coronavirus 2 (SARS-CoV-2) spike glycopro-
tein, and pharmaceutical compositions containing the same.**20 Claims, 9 Drawing Sheets****Specification includes a Sequence Listing.**

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FIG. 1

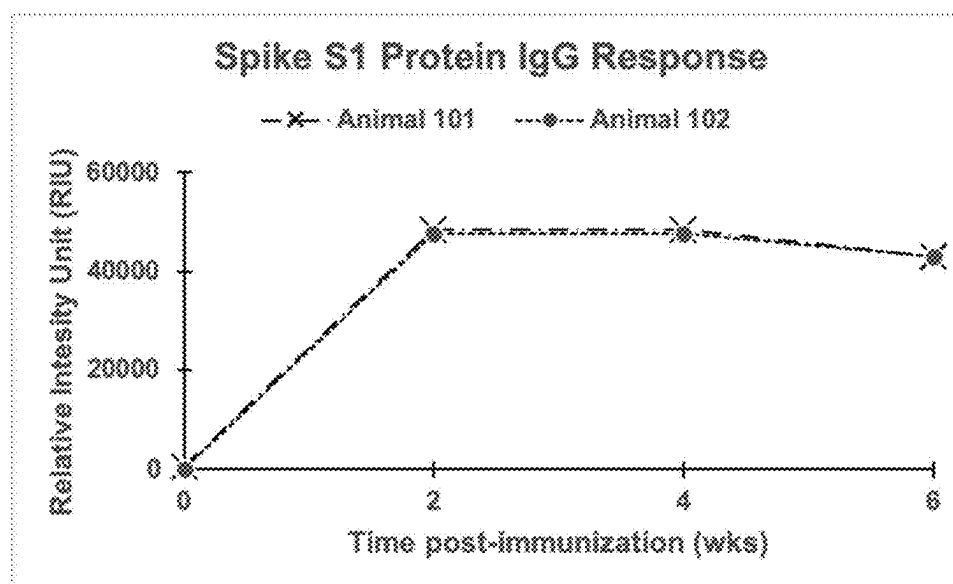


FIG. 2A

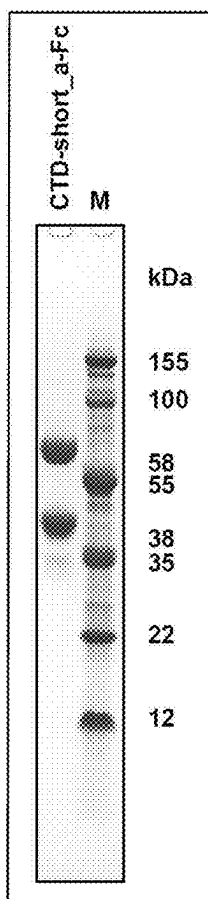


FIG. 2B

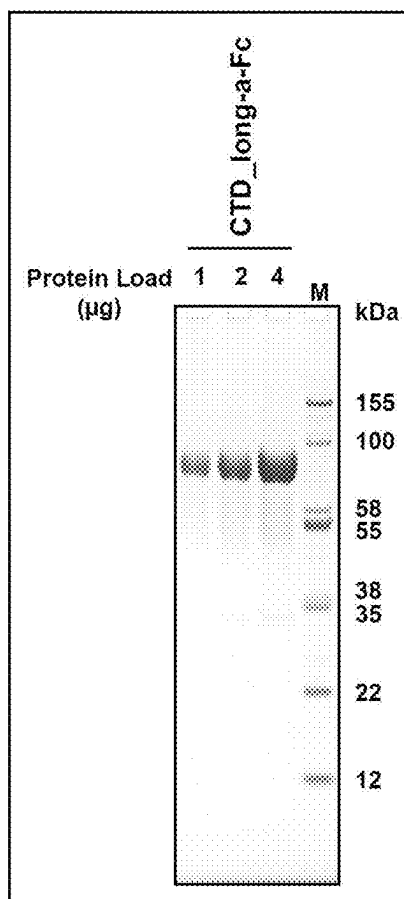


FIG. 2C

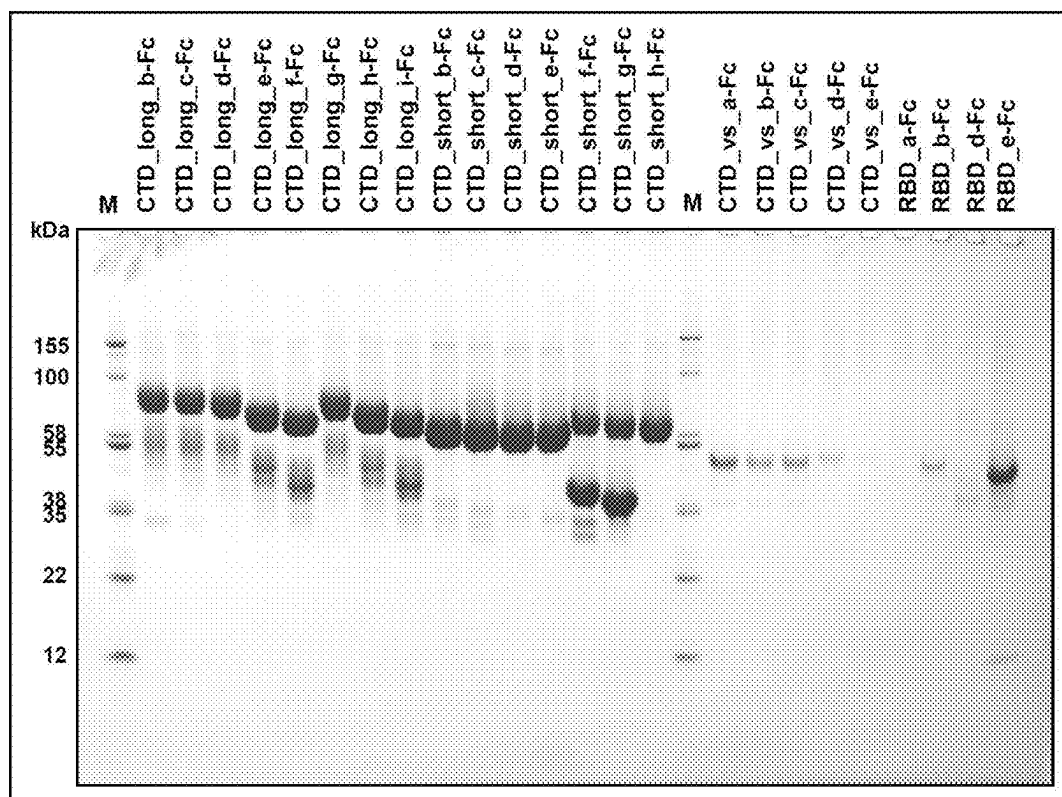


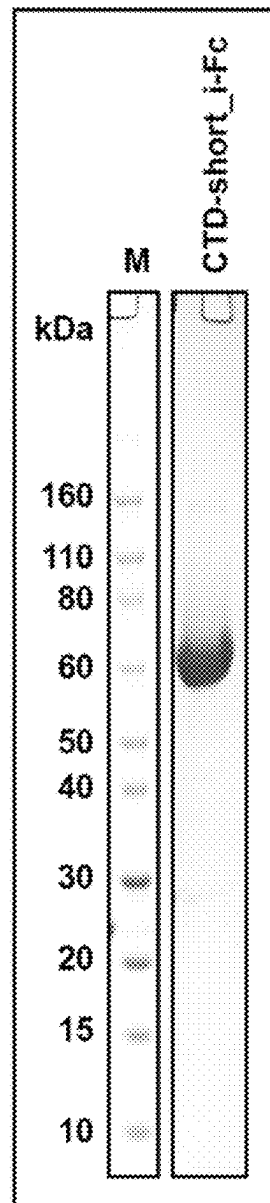
FIG. 2D

FIG. 3

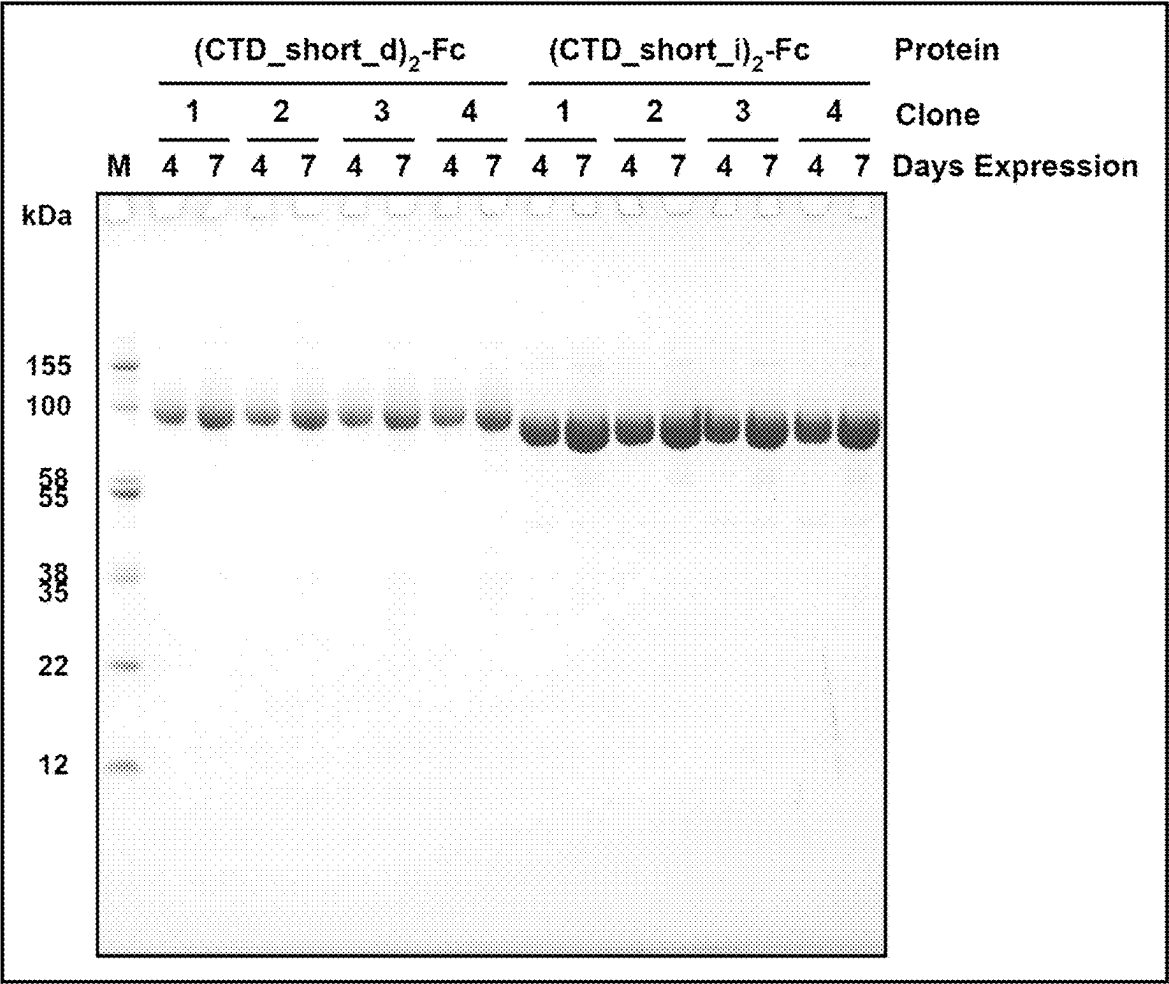


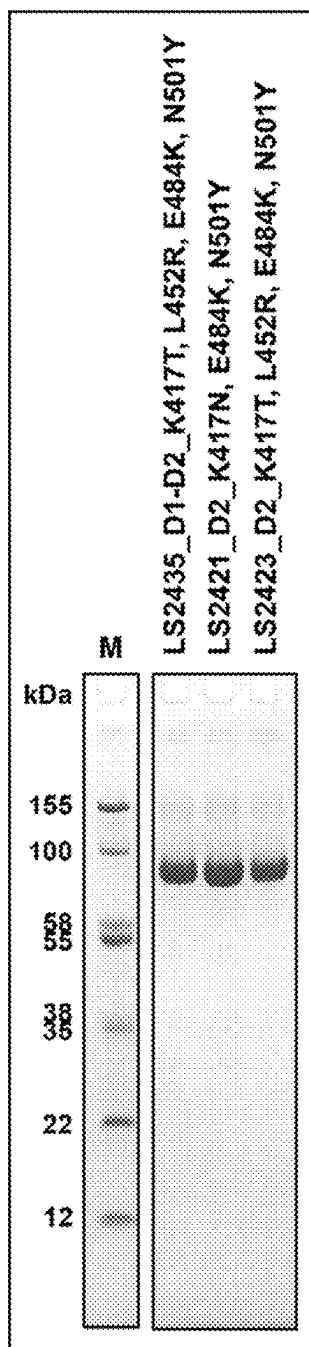
FIG. 4

FIG. 5A

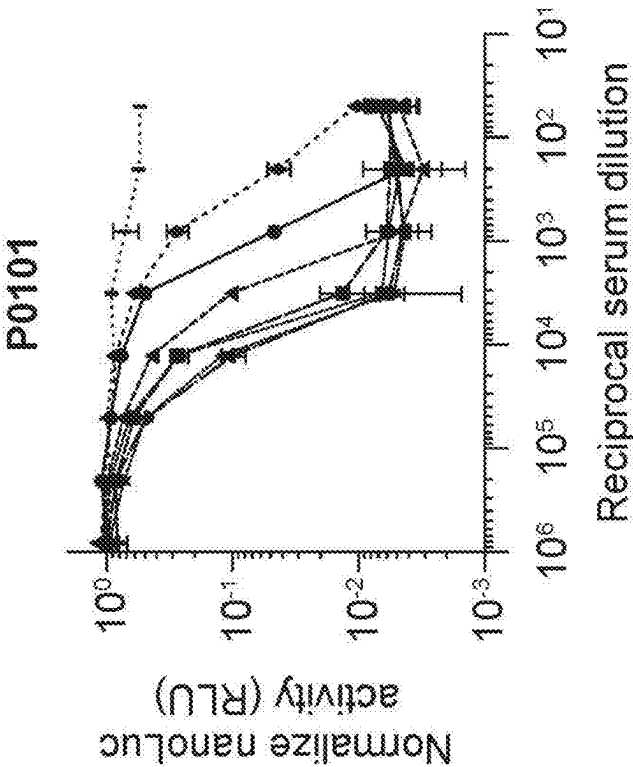


FIG. 5B

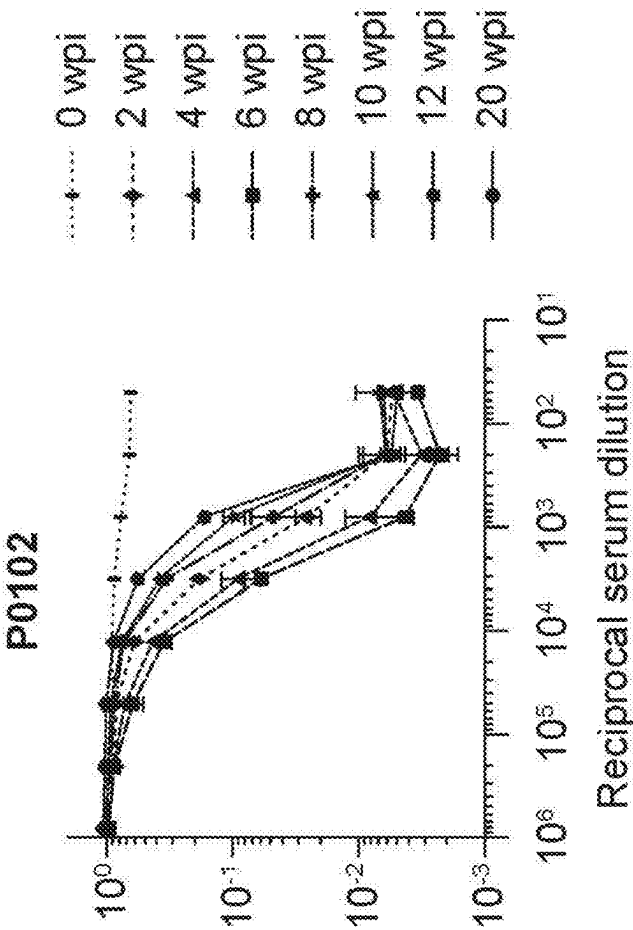


FIG. 5C

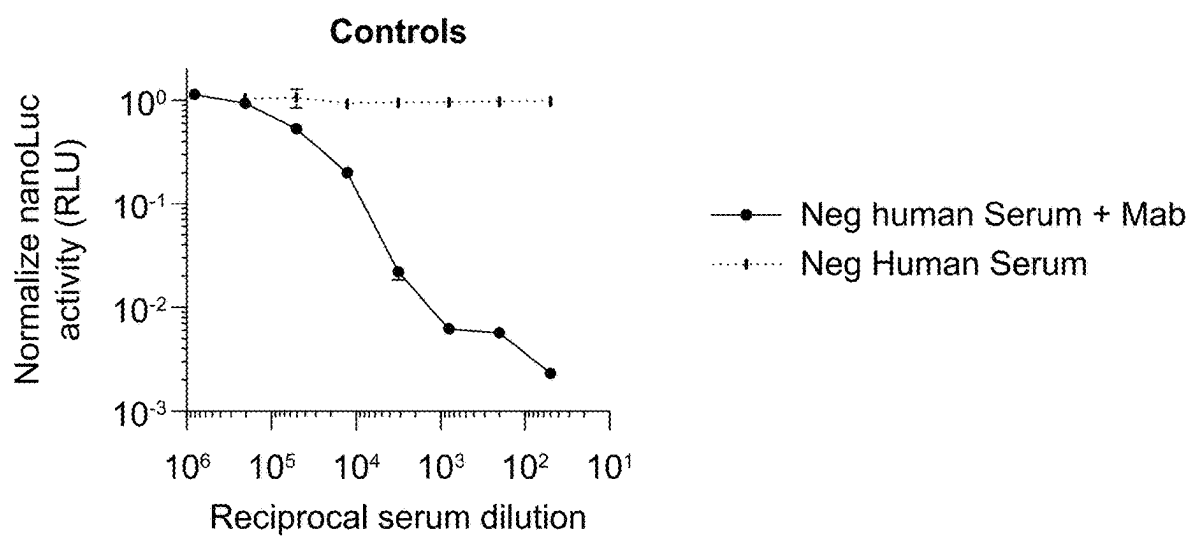
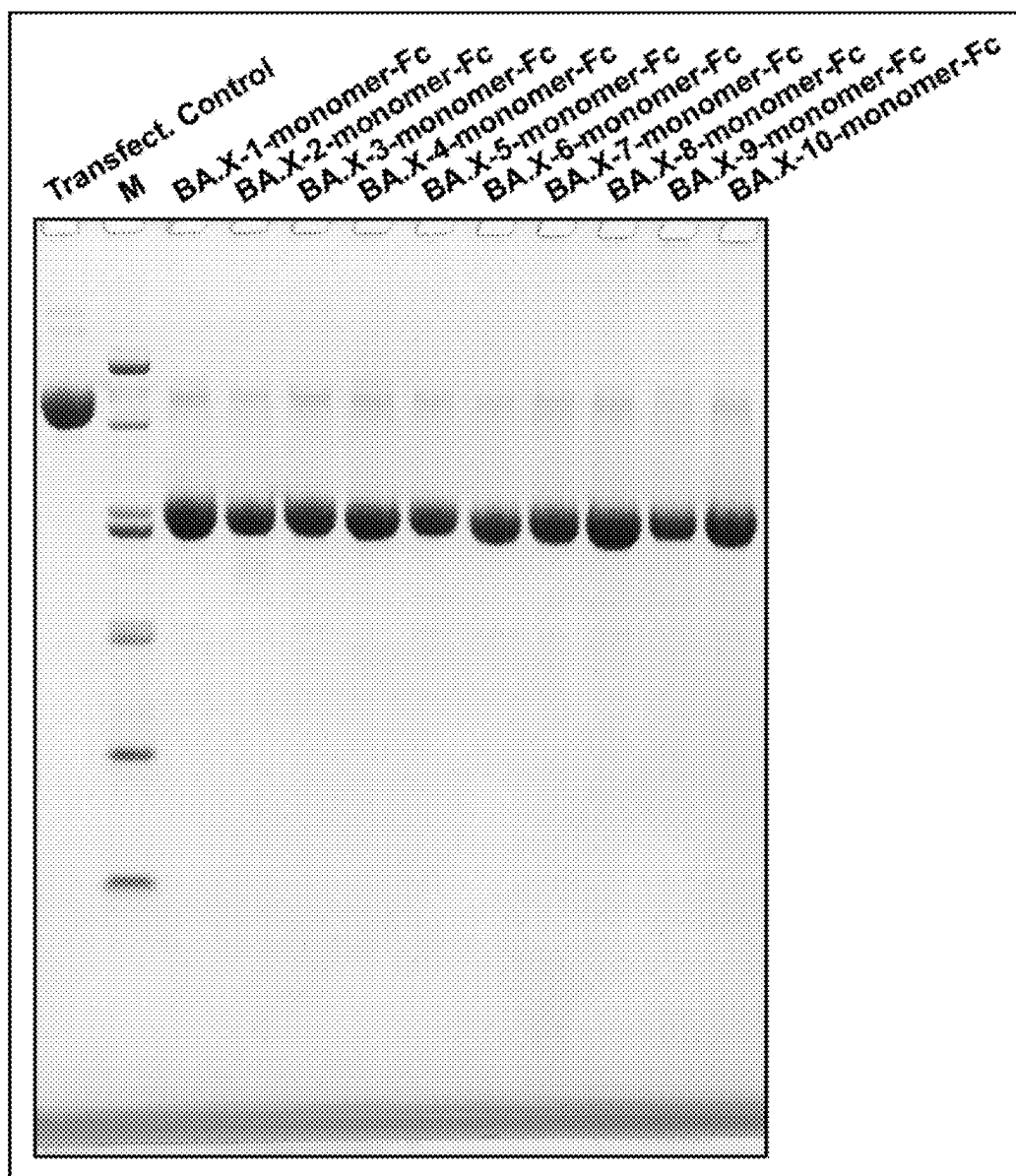


FIG. 6



RECOMBINANT POLYPEPTIDES CONTAINING AT LEAST ONE IMMUNOGENIC FRAGMENT AND USES THEREOF

CROSS-REFERENCE TO PRIOR APPLICATION

This application claims benefit to pending U.S. patent application Ser. No. 17/535,309, filed Nov. 24, 2021, which is a continuation application of International Patent Application No. PCT/US2021/040019, filed Jun. 30, 2021, which claims benefit to U.S. Provisional Patent Applications No. 63/046,426, filed Jun. 30, 2020, and 63/154,647, filed Feb. 26, 2021, all of which are each hereby incorporated by reference in their entireties.

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ELECTRONICALLY

Incorporated by reference in its entirety herein is a computer-readable nucleotide/amino acid sequence listing submitted concurrently herewith and identified as follows: One 848,455Byte ASCII (Text) file named "759990_ST25.txt," created on Apr. 15, 2022.

BACKGROUND OF THE INVENTION

The rapid evolution of new SARS-CoV-2 variants containing mutations that alter the amino acid sequence of the Spike protein resulting in altered function, and altered resistance to native immune defenses and to immune defenses elicited by currently marketed vaccines has led to a need for alternative platforms that allow for incorporation of new variant mutations in a robust vaccine.

BRIEF SUMMARY OF THE INVENTION

The present invention relates to recombinant polypeptides that include at least one immunogenic fragment of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein and, optionally, an antibody Fc region. In some embodiments, the recombinant polypeptide includes more than one immunogenic fragment, e.g., two, three, four, five, or more immunogenic fragments. In some embodiments, the recombinant polypeptide includes one or more immunogenic fragments of Severe Acute Respiratory Syndrome Coronavirus (SARS-CoV) and/or Middle Eastern Respiratory Syndrome Coronavirus (MERS-CoV) and optionally an antibody Fc region, optionally in combination with one or more SARS-CoV-2 immunogenic fragments.

The present invention further relates to pharmaceutical compositions, such as vaccines, that include the recombinant polypeptide. In some embodiments, the pharmaceutical composition includes an adjuvant.

The present invention also relates to a method for preventing, inhibiting, reducing, eliminating, protecting, and/or delaying the onset of an infection or an infectious clinical condition caused by a beta coronavirus in a subject, wherein the method includes administering to the subject at least one recombinant polypeptide of the invention or a pharmaceutical composition including the same.

The present invention further relates to a method for inducing an immune response against a coronavirus in a subject, wherein the method includes administering to the subject at least one recombinant polypeptide of the invention or a pharmaceutical composition including the same.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a graph depicting Spike 51 protein IgG response in Rhesus macaques at certain time points (in weeks) after initial injection with an exemplary construct.

FIG. 2A is an image showing CHO cell expression of CTD_short_a-Fc (LS2330) after 5 days. Protein was affinity purified using Protein A agarose and analyzed by reducing SDS-PAGE and detected by Coomassie R-250 staining. M-Molecular weight markers. As can be seen in each of FIGS. 2A-2D, the constructs were resistant to proteolytic degradation during expression thus facilitating higher yields of intact, soluble protein.

FIG. 2B is an image showing CHO cell expression of CTD_long_a-Fc (LS3472) after 5 days. Protein was affinity purified using Protein A agarose and analyzed by reducing SDS-PAGE and detected by Coomassie R-250 staining. Protein loads per lane are indicated.

FIG. 2C is an image showing CHO cell expression of CTD-deletion series after 5 days. Protein was affinity purified using Protein A agarose from equal volumes of transfected culture. Eluted protein was loaded based on volume, separated by reducing SDS-PAGE, and detected by Coomassie R-250 staining. Load volumes were twice as large for samples (left to right) CTD_vs_a-Fc to RBD_e-Fc compared to the samples on the left of the gel (CTD_long_b-Fc to CTD_short_h-Fc).

FIG. 2D is an image showing CHO cell expression of CTD_short_i-Fc (LS2371) after 5 days. Protein was affinity purified using Protein A agarose and analyzed by reducing SDS-PAGE and detected by Coomassie R-250 staining.

FIG. 3 is an image showing CHO cell expression of (CTD_short_d)₂-Fc (clone 1 through 4 correspond to strains LS2397 through 2400; SEQ ID NO: 161) and (CTD_short_i)₂-Fc (clone 1 through 4 correspond to strains LS2401 through 2404; SEQ ID NO: 163) after 4 or 7 days as indicated. Protein was affinity purified using Protein A agarose from equal volumes of transfected culture. Eluted protein was loaded based on volume, separated by reducing SDS-PAGE, and detected by Coomassie R-250 staining. M-Molecular weight markers.

FIG. 4 is an image showing CHO cell expression of mutant (CTD_short_i)₂-Fc constructs wherein amino acid mutations corresponding to newly identified SARS-CoV-2 variants have been added to both CTD domains (D1 and D2) or second domain (D2) only of the CTD dimer. The mutants tested include (a) D1 and D2 mutations for hybrid P.1 and CAL.20C variants; K417T, L452R, E484K, N501Y (Strain 2435), (b) D2 mutations for 501.V2 variant/B.1.351, K417N, E484K, N501Y (Strains 2421), and (c) D2 mutations for hybrid P.1 and CAL.20C variants; K417T, L452R, E484K, N501Y (Strain 2423). Protein was affinity purified using Protein A agarose from equal volumes of 4-day transfected culture. Eluted protein was loaded based on volume, separated by reducing SDS-PAGE, and detected by Coomassie R-250 staining. M-Molecular weight markers. Protein loads were 3.48 (strain 2435), 4.52 (strain 2421), and 3.14 µg (strain 2423).

FIGS. 5A, 5B, and 5C are graphs showing the results of neutralization assays using serum samples from animals P0101 (FIG. 5A) and P0102 (FIG. 5B) immunized with SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]), which were the subject of analysis discussed in Example 1. FIG. 5C represents control samples including human serum from an individual prior to the emergence of SARS-CoV-2 (negative human serum sample) and the same serum spiked with a monoclonal antibody immunoreactive to the RBD portion of

the Wuhan variant of the SARS-CoV-2 Spike protein. Infectivity of serum neutralized pseudotypes virus using 295T/ACE2 target cells was quantified by measuring NanoLuc luciferase activity (RLU) and graphed on the y-axis. Reciprocal serum dilution is shown on the x-axis.

FIG. 6 is an image showing CHO cell expression of BA.X-1-monomer-Fc-BA.X-10-monomer-Fc after 4 days. Protein was affinity purified using Protein A agarose from equal volumes of transfected culture. Eluted protein was loaded based on volume, separated by reducing SDS-PAGE, and detected by Coomassie R-250 staining. M-Molecular weight markers. 'Transfect. Control' is an ACE2-Fc transfection, expression control.

DETAILED DESCRIPTION OF THE INVENTION

Recombinant polypeptides of the invention can include any suitable immunogenic fragment or fragments of the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein and optionally any suitable antibody Fc region. In some embodiments, an immunogenic fragment comprises, consist of, or consist essentially of, the N-terminal domain of the S1 subunit, the C-terminal domain of the S1 subunit, or both. In certain embodiments, an immunogenic fragment can include the complete SARS-CoV-2 spike glycoprotein. Recombinant polypeptides of the invention include at least one immunogenic fragment, and can contain two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, or more such immunogenic fragments.

In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof. Wild-type spike glycoproteins include those of any SARS-CoV-2 strain that has been isolated from a subject. Examples include Wuhan-Hu-1, VOC 202012/01/B.1.1.7 (Alpha or UK), VOC-202102/02 (B.1.1.7 with E484K) (UK), 501.V2/B.1.351 (South Africa), B.1.429/CAL.20C (California), B.1.525, and Lineage P.1 (Gamma or Brazil), B.1.427 (Epsilon), B.1.429 (Epsilon), B.1.617.1, B.1.617.2 (Delta), B.1.526 (Iota), B.1.617.3, B.1.A.2.5, C.36.3, B.1.1.318, B.1.351, B.1.621, B.1.525, P.1.1, P.2 (Zeta), B.1.623, R.1, B.1.1.7, B.1.351, B.1.351.3, B.1.351.3, BA.1, BA.1.1, BA.1.1.1, BA.1.1.2, BA.1.1.12, BA.1.1.13, BA.1.14, BA.1.15, BA.1.15.1, BA.1.16, BA.1.17.2, BA.1.18, BA.2, BA.2+, BA.2.1, BA.2.10.1, BA.2.2, BA.2.3.2, BA.2.4, BA.2.5, BA.2.6, BA.2.7, BA.2.8, BA.2.9, BA.2.10, BA.2.12, BA.3, BA.4, BA.5, XA, XB, XC, XD, XE, XF, XG, XH, XJ, XK, XL, XM, XN, XP, XQ, XR, XS, and XT. In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV or MERS spike glycoprotein, or any portion thereof. Wild-type SARS-CoV and MERS spike glycoproteins include those of any SARS-CoV or MERS strain that has been isolated from a subject. Examples of such strains include SARS coronavirus Tor2 (GenBank accession number NC_004718.3) and MERS coronavirus (GenBank accession number NC_019843.3).

In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof. In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%,

sequence identity to a wild-type SARS-CoV or MERS spike glycoprotein, or any portion thereof.

In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof, except for at one or more of the following positions of the amino acid sequence: L5, A67, H69, V70, D80, T95, G142, Y144, E154, F157, D253, K417, L452, S477, T478, E484, N501, D614, Q677, P681, A701, T791, T859, F888, D950, and Q1071, wherein the positions of the listed amino acid residues correspond to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof, except for at one or more of the following positions of the amino acid sequence: L5, A67, H69, V70, D80, T95, G142, Y144, E154, F157, D253, G339, R346, S371, S373, S375, T376, D405, R408, K417, N440, G446, L452, S477, T478, E484, Q493, G496, Q498, N501, Y505, T547, D614, Q677, P681, A701, T791, T859, F888, D950, and Q1071, wherein the positions of the listed amino acid residues correspond to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). In certain embodiments, one or more of the immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence selected from SEQ ID NOs: 245-254, except for at one or more of the following positions of the amino acid sequence: L5, A67, H69, V70, D80, T95, G142, Y144, E154, F157, D253, G339, R346, S371, S373, S375, T376, D405, R408, K417, N440, G446, L452, S477, T478, E484, Q493, G496, Q498, N501, Y505, T547, D614, Q677, P681, A701, T791, T859, F888, D950, and Q1071, wherein the positions of the listed amino acid residues correspond to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). In certain embodiments, one or more of the immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence selected from SEQ ID NOs: 245-254, except for at one or more of the following positions of the amino acid sequence: G339, R346, S371, S373, S375, T376, D405, R408, K417, N440, G446, L452, S477, T478, E484, Q493, G496, Q498, N501, and Y505, wherein the positions of the listed amino acid residues correspond to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). The mutations at these positions can be any suitable mutation, including conservative and non-conservative amino acid mutations. For instance, a conservative substitution can replace one aliphatic amino acid (i.e., Glycine, Alanine, Valine, Leucine, Methionine or Isoleucine) for another, one polar, uncharged R group amino acid (i.e., Serine, Cysteine, Threonine, Proline, Asparagine, or Methionine) for another, one positively charged R group amino acid (i.e., Histidine, Lysine, or Arginine) for another, one negatively charged R group amino acid (i.e., Aspartate or Glutamate) for another, and one non-polar, aromatic R group amino acid (i.e., Phenylalanine, Tyrosine, or Tryptophan) for another. In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof, except for one or more of the following amino acid substitutions and deletions: L5F, A67V, 69del, 70del, D80G, T95I, G142D, I44del, E154K, F157S, D253G, L452R, S477N, E484K, E484Q, K417N, K417T, T478K, N501Y, D614G, Q677H, T791I, P681H, P681R, A701V, F888L, T859N, D950H, D950N, and Q1071H, wherein the listed amino acid substitutions and deletions are relative to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/

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QHD43416). In one embodiment, the one or more amino acid substitutions and/or deletions is L452R. In another embodiment, the one or more amino acid substitutions and/or deletions is E484K. In a further embodiment, the one or more protein substitutions and/or deletions are K417N, E484K, and N501Y. In yet another embodiment, the one or more substitutions are and/or deletions are K417T, E484K, and N501Y. In another embodiment, the one or more amino acid substitutions and/or deletions are N501Y, 69del, 70del, and P681H. In another embodiment, the one or more amino acid substitutions and/or deletions are K417T, L452R, E484K, and N501Y.

In certain embodiments, one or more of the immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence comprising, consisting of, or consisting essentially of a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof, wherein the amino acid sequence further comprises one or more of the following amino acid substitutions: G339D, R346K, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, G446S, L452R, S477N, T478K, E484A, E484K, Q493R, G496S, Q498R, N501Y, and Y505H to the extent the sequence does not already contain such residues. In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof, except for one or more of the following amino acid substitutions and deletions: L5F, A67V, 69del, 70del, D80G, T95I, G142D, 144del, E154K, F157S, D253G, G339D, R346K, S371F, S373P, S375F, T376A, D405N, R408S, N440K, G446S, L452R, S477N, E484A, E484K, E484Q, K417N, K417T, T478K, Q493R, G496S, Q498R, N501Y, Y505H, T547K, D614G, Q677H, T791I, P681H, P681R, A701V, F888L, T859N, D950H, D950N, and Q1071H, wherein the listed amino acid substitutions and deletions are relative to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416).

In some embodiments, one or more of the immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence selected from SEQ ID NOs: 245-254, except for one or more of the following amino acid substitutions: G339D, R346K, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, G446S, L452R, S477N, T478K, E484A, E484K, Q493R, G496S, Q498R, N501Y, and Y505H, wherein the positions of the listed amino acid residues correspond to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). In certain embodiments, one or more of the immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from SEQ ID NOs: 245-254, wherein the amino acid sequence further comprises one or more of the following amino acid substitutions: G339D, R346K, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, G446S, L452R, S477N, T478K, E484A, E484K, Q493R, G496S, Q498R, N501Y, and Y505H, to the extent the sequence does not already contain such residues.

In one embodiment, the one or more amino acid substitutions is G339D, R346K, T376A, D405N, R408S, K417N, N440K, L452R, S477N, T478K, E484A, Q493R, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, S373P, T376A, D405N, R408S, K417N, N440K, S477N, T478K, E484A, Q493R, Q498R, N501Y, and Y505H. In another embodi-

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ment, the one or more amino acid substitutions is G339D, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, S477N, T478K, E484A, Q493R, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, T376A, D405N, R408S, K417N, N440K, L452R, S477N, T478K, E484K, Q493R, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, L452R, S477N, T478K, E484K, Q493R, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, S477N, T478K, E484A, and N501Y. In another embodiment, the one or more amino acid substitutions is G339D, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, S477N, T478K, E484A, Q493R, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, S371F, K417N, N440K, G446S, S477N, T478K, E484A, Q493R, G496S, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, R346K, T376A, D405N, R408S, K417N, N440K, G446S, L452R, S477N, T478K, E484K, Q493R, G496S, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, S373P, T376A, D405N, R408S, K417N, N440K, S477N, T478K, E484A, Q493R, Q498R, N501Y, and Y505H.

The subject can be mammalian, including human, non-human primate, horse, pig, cattle, cat, dog, sheep, mink, rodent, hamster, or bat. Subjects can further include western lowland gorilla, northern white-cheeked gibbon, Sumatran orangutan, crab-eating macaque, drill, proboscis monkey, bonobo, chimpanzee, ugandan red colobus, red-shanked douc, golden snub-nosed monkey, green monkey, patas monkey, rhesus macaque, olive baboon, gelade, sooty mangabey, southern pig-tailed macaque, angola colobus, coquerel's sifaka, Gambian pouched rat, Chinese hamster, common gund, beluga whale, blue-eyed black lemur, indri, narwhal, narrow-ridged finless porpoise, harbour porpoise, minke whale, Antarctic minke whale, gray whale, spalax, white-tailed deer, reindeer, southern tamandua, Stephens's kangaroo rat, Pere David's deer, transcaucasian mole vole, long-finned pilot whale, Pacific white-sided dolphin, baiji, giant anteater, muskrat, killer whale, common bottlenose dolphin, aye-aye, fat-tailed dwarf lemur, thirteen-lined ground squirrel, yellow-bellied marmot, alpine marmot, golden hamster, sperm whale, daurian ground squirrel, gobi jerboa, barbary sheep, pronghorn, Nancy ma's night monkey, hirola, American bison, zebu, wild yak, cattle, water buffalo, white-eared titi, common marmoset, wild goat, goat, Panamanian white-faced capuchin, cat, Masai giraffe, Nilgiri tahr, Candian lynx, coquerel's giant mouse lemur, Siberian musk deer, sunda clouded leopard, clouded leopard, okapi, sheep, jaguar, leopard, Siberian tiger, Tibetan antelope, little pocket mouse, deer mouse, white-faced saki, cougar, black-capped squirrel monkey, tufted capuchin, arctic ground squirrel, *bos indicus* × *bos Taurus*, cheetah, mantled howler, Geoffroy's spider monkey, Damaraland mole-rat, naked mole-rat, hippopotamus, snowshoe hare, Dama gazelle, European rabbit, scimitar oryx, emperor tamarin, and alpaca.

In some embodiments, one or more of the immunogenic fragments contain one or more mutations, such that the one or more immunogenic fragments are not identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof. The one or more mutations can be any suitable mutation and/or deletion, such as those disclosed herein. For example, an immunogenic fragment can contain sequences

from two, three, four, five, six, seven, eight, nine, ten, or more strains, such that the resulting fragment is no longer identical to any of its parent strains. In this way, a single immunogenic fragment can present epitopes from multiple wild-type SARS-CoV-2 spike glycoproteins, or any portion thereof. This can lead to a more robust immune response and increased immune protection from a range of SARS-CoV-2 strains in a subject when a recombinant polypeptide of the invention containing one or more such immunogenic fragments, or a pharmaceutical composition containing the same, is administered to the subject.

In some embodiments, the nucleic acid sequence encoding an immunogenic fragment includes one, two, three, four, five, six, seven, eight, nine, ten, twenty, thirty, forty, or more point mutations and/or deletions in comparison to the nucleic acid sequence encoding the corresponding fragment of a wild-type or mutant glycoprotein. In some embodiments, the amino acid sequence of an immunogenic fragment includes one, two, three, four, five, six, seven, eight, nine, ten, twenty, thirty, forty, or more substitutions and/or deletions in comparison to the amino acid sequence encoding corresponding fragment of a wild-type or mutant glycoprotein.

In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 1 (CTD_long_a), SEQ ID NO: 3 (CTD_long_a D614G), SEQ ID NO: 5 (CTD_long_a-Dimer), SEQ ID NO: 7 (CTD_long_b), SEQ ID NO: 9 (CTD_long_c), SEQ ID NO: 11 (CTD_long_d), SEQ ID NO: 13 (CTD_long_e), SEQ ID NO: 15 (CTD_long_f), SEQ ID NO: 17 (CTD_long_g), SEQ ID NO: 19 (CTD_long_h), SEQ ID NO: 21 (CTD_long_i), SEQ ID NO: 23 (CTD_short_a), SEQ ID NO: 25 (CTD_short_b), SEQ ID NO: 27 (CTD_short_c), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 33 (CTD_short_f), SEQ ID NO: 35 (CTD_short_g), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 41 (CTD_vs_a), SEQ ID NO: 43 (CTD_vs_b), SEQ ID NO: 45 (CTD_vs_c), SEQ ID NO: 47 (CTD_vs_d), SEQ ID NO: 49 (CTD_vs_e), SEQ ID NO: 51 (RBD_a), SEQ ID NO: 53 (RBD_b), SEQ ID NO: 55 (RBD_c), SEQ ID NO: 57 (RBD_d), SEQ ID NO: 59 (RBD_e), SEQ ID NO: 61 (NTD_long_a), SEQ ID NO: 63 (NTD_short_a), SEQ ID NO: 171 (RBD-tight), SEQ ID NO: 173 ((RBD-tight)₂), SEQ ID NO: 175 (RBD-extended), SEQ ID NO: 177 ((RBD-extended)₂), SEQ ID NO: 179 (RBD), SEQ ID NO: 181 ((RBD)₂), SEQ ID NO: 183 ((CTD_short_d)₂), SEQ ID NO: 185 ((CTD_short_i)₂), SEQ ID NO: 187 ((CTD_short_i)₂—mod. 1), SEQ ID NO: 189 ((CTD_short_i)₂—mod. 2), SEQ ID NO: 191 ((CTD_short_i)₂—mod. 3), SEQ ID NO: 199 (SARS-2003, SARS_short_h), SEQ ID NO: 201 (SARS-2003, SARS_short_i), SEQ ID NO: 203 (MERS_Lytic_a), SEQ ID NO: 205 (MERS_Lytic_b), SEQ ID NO: 207 (MERS_Lytic_c), SEQ ID NO: 209 (MERS_Lytic_d), SEQ ID NO: 211 (MERS_Lytic_e), SEQ ID NO: 213 (MERS_Lytic_f), SEQ ID NO: 215 (MERS_Lytic_g), SEQ ID NOS: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOS: 265-274 (BA.X-1-dimer-BA.X-10-dimer). In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 185 ((CTD_short_i)₂),

SEQ ID NO: 187 ((CTD_short_i)₂—mod. 1), SEQ ID NO: 189 ((CTD_short_i)₂—mod. 2), and SEQ ID NO: 191 ((CTD_short_i)₂—mod. 3). In further embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence of SEQ ID NO: 185 ((CTD_short_i)₂). In further embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 183 ((CTD_short_d)₂), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 37 (CTD_short_h), and SEQ ID NO: 23 (CTD_short_a). In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 39. In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 185. In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 190. In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence selected from SEQ ID NOS: 245-254 (BA.X-1-BA.X-10). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 245 (BA.X-1). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 246 (BA.X-2). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 247 (BA.X-3). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 248 (BA.X-4). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 249 (BA.X-5). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino

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In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NOS: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 171, 173, 175, 177, 179, 181, 183, 185, 187, 189, 191, 199, 201, 203, 205, 207, 209, 211, 213, 215, 245-254, and 265-274. In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NO: 185 ((CTD_short_i)₂), SEQ ID NO: 187 ((CTD_short_i)₂—mod. 1), SEQ ID NO: 189 ((CTD_short_i)₂—mod. 2), and SEQ ID NO: 191 ((CTD_short_i)₂—mod. 3). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of the amino acid sequence represented by SEQ ID NO: 185. In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of the amino acid sequence represented by SEQ ID NO: 191. In further embodiments an immunogenic fragment comprises, consists of, or consists essentially of the amino acid sequence of SEQ ID NO: 185 ((CTD_short_i)₂). In further embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NO: 183 ((CTD_short_d)₂), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 37 (CTD_short_h), and SEQ ID NO: 23 (CTD_short_a). In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NOS: 245-254 and 265-274. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NOS: 245-254. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 245. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 246. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 247. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 248. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 249. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 250. In some

embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 251. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 252. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 253. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 254. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 265. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 266. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 267. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 268. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 269. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 270. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 271. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 272. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 273. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence represented by SEQ ID NO: 274.

In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that is encoded by a nucleotide sequence that comprises, consists of, or consists essentially of a nucleotide sequence selected from the group consisting of SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 172, 174, 176, 178, 180, 182, 184, 186, 188, 190, 192, 200, 202, 204, 206, 208, 210, 212, 214, 216, 285-294, and 305-314. In some embodiments, the recombinant polypeptide includes at least two immunogenic fragments. For example, the recombinant polypeptide can include two, three, four, five, six, seven, eight, nine, ten, or more immunogenic fragments. In some embodiments, the recombinant polypeptide includes at least three immunogenic fragments. In some embodiments, the recombinant

polypeptide includes at least three immunogenic fragments, of which at least one is a SARS-CoV-2 spike glycoprotein fragment as described herein, at least one is a SARS-CoV spike glycoprotein fragment as described herein, and at least one is a MERS-CoV spike glycoprotein fragment as described herein. In some embodiments, the recombinant polypeptide includes at least three immunogenic fragments, of which at least one is a SARS-CoV-2 spike glycoprotein fragment, wherein the SARS-CoV-2 fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 1 (CTD_long_a), SEQ ID NO: 3 (CTD_long_a D614G), SEQ ID NO: 5 (CTD_long_a-Dimer), SEQ ID NO: 7 (CTD_long_b), SEQ ID NO: 9 (CTD_long_c), SEQ ID NO: 11 (CTD_long_d), SEQ ID NO: 13 (CTD_long_e), SEQ ID NO: 15 (CTD_long_f), SEQ ID NO: 17 (CTD_long_g), SEQ ID NO: 19 (CTD_long_h), SEQ ID NO: 21 (CTD_long_i), SEQ ID NO: 23 (CTD_short_a), SEQ ID NO: 25 (CTD_short_b), SEQ ID NO: 27 (CTD_short_c), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 33 (CTD_short_f), SEQ ID NO: 35 (CTD_short_g), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 41 (CTD_vs_a), SEQ ID NO: 43 (CTD_vs_b), SEQ ID NO: 45 (CTD_vs_c), SEQ ID NO: 47 (CTD_vs_d), SEQ ID NO: 49 (CTD_vs_e), SEQ ID NO: 51 (RBD_a), SEQ ID NO: 53 (RBD_b), SEQ ID NO: 55 (RBD_c), SEQ ID NO: 57 (RBD_d), SEQ ID NO: 59 (RBD_e), SEQ ID NO: 61 (NTD_long_a), SEQ ID NO: 63 (NTD_short_a), SEQ ID NO: 171 (RBD-tight), SEQ ID NO: 173 ((RBD-tight)₂), SEQ ID NO: 175 (RBD-extended), SEQ ID NO: 177 ((RBD-extended)₂), SEQ ID NO: 179 (RBD), SEQ ID NO: 181 ((RBD)₂), SEQ ID NO: 183 ((CTD_short_d)₂), SEQ ID NO: 185 ((CTD_short_i)₂), SEQ ID NO: 187 ((CTD_short_i)₂—mod. 1), SEQ ID NO: 189 ((CTD_short_i)₂—mod. 2), SEQ ID NO: 191 ((CTD_short_i)₂—mod. 3), SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOs: 265-274 (BA.X-1-dimer-BA.X-10-dimer); at least one is a SARS-CoV spike glycoprotein fragment, wherein the SARS-CoV fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence represented by SEQ ID NO: 199 (SARS-2003, SARS_short_h), or SEQ ID NO: 201 (SARS-2003, SARS_short_i); and at least one is a MERS-CoV spike glycoprotein fragment, wherein the MERS-CoV fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 203 (MERS_Lytic_a), SEQ ID NO: 205 (MERS_Lytic_b), SEQ ID NO: 207 (MERS_Lytic_c), SEQ ID NO: 209 (MERS_Lytic_d), SEQ ID NO: 211 (MERS_Lytic_e), SEQ ID NO: 213 (MERS_Lytic_f), and SEQ ID NO: 215 (MERS_Lytic_g).

In some embodiments, the recombinant polypeptide includes at least two immunogenic fragments, wherein each of the at least two immunogenic fragments comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence independently selected from the group consisting of SEQ ID NO: 1 (CTD_long_a), SEQ ID NO: 3 (CTD_long_a D614G), SEQ ID NO: 5 (CTD_long_a-Dimer), SEQ ID NO: 7 (CTD_long_b), SEQ ID NO: 9

(CTD_long_c), SEQ ID NO: 11 (CTD_long_d), SEQ ID NO: 13 (CTD_long_e), SEQ ID NO: 15 (CTD_long_f), SEQ ID NO: 17 (CTD_long_g), SEQ ID NO: 19 (CTD_long_h), SEQ ID NO: 21 (CTD_long_i), SEQ ID NO: 23 (CTD_short_a), SEQ ID NO: 25 (CTD_short_b), SEQ ID NO: 27 (CTD_short_c), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 33 (CTD_short_f), SEQ ID NO: 35 (CTD_short_g), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 41 (CTD_vs_a), SEQ ID NO: 43 (CTD_vs_b), SEQ ID NO: 45 (CTD_vs_c), SEQ ID NO: 47 (CTD_vs_d), SEQ ID NO: 49 (CTD_vs_e), SEQ ID NO: 51 (RBD_a), SEQ ID NO: 53 (RBD_b), SEQ ID NO: 55 (RBD_c), SEQ ID NO: 57 (RBD_d), SEQ ID NO: 59 (RBD_e), SEQ ID NO: 61 (NTD_long_a), SEQ ID NO: 63 (NTD_short_a), SEQ ID NO: 171 (RBD-tight), SEQ ID NO: 173 ((RBD-tight)₂), SEQ ID NO: 175 (RBD-extended), SEQ ID NO: 177 ((RBD-extended)₂), SEQ ID NO: 179 (RBD), SEQ ID NO: 181 ((RBD)₂), SEQ ID NO: 183 ((CTD_short_d)₂), SEQ ID NO: 185 ((CTD_short_i)₂), SEQ ID NO: 187 ((CTD_short_i)₂-mod. 1), SEQ ID NO: 189 ((CTD_short_i)₂-mod. 2), SEQ ID NO: 191 ((CTD_short_i)₂-mod. 3), SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOs: 265-274 (BA.X-1-dimer-BA.X-10-dimer). In some embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence independently selected from the group consisting of SEQ ID NO: 183 ((CTD_short_d)₂), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 23 (CTD_short_a), and SEQ ID NOs: 245-254 (BA.X-1-BA.X-10). In some embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence of SEQ ID NO: 39 (CTD_short_i). In further embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence independently selected from the group consisting of SEQ ID NO: 183 ((CTD_short_d)₂), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 23 (CTD_short_a), and SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOs: 265-274 (BA.X-1-dimer-BA.X-10-dimer). In further embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of the amino acid sequence of SEQ ID NO: 39 (CTD_short_i). In further embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence independently selected from the group consisting of SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOs: 265-274 (BA.X-1-dimer-BA.X-10-dimer). In further embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence independently selected from the group consisting of SEQ ID NOs: 245-254 (BA.X-1-BA.X-10). The at least two immunogenic fragments can comprise, consist of, or consist essentially of, any suitable combination of amino acid sequences.

In certain embodiments, each of the at least two immunogenic fragments comprise, consist of, or consist essentially of the same amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254 (BA.X-1-BA.X-

10), e.g., the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 245. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of SEQ ID NO: 246. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 247. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 248. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 249. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 250. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 251. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 252. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 253. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 254. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 265. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 266. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 267. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 268. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 269. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 270. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 271. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 272. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 273. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 274.

In certain embodiments, each of the at least two immunogenic fragments comprise, consist of, or consist essentially of a different amino acid sequence independently selected from the group consisting of SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), e.g., when the recombinant polypeptide comprises, consists of, or consists essentially of two immunogenic fragments, one immunogenic fragment comprises, consists of, or consists essentially of SEQ ID NO: 245 and

the second immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NOs: 246-254.

In some embodiments, the recombinant polypeptide includes a plurality of identical immunogenic fragments. For example, a recombinant polypeptide can include two, three, four, five, six, seven, eight, nine, ten, or more immunogenic fragments, wherein each fragment comprises, consists of, or consists essentially of the same amino acid sequence. In some embodiments, the recombinant polypeptide includes two, three, four or five identical immunogenic fragments. In yet further embodiments, the recombinant polypeptide includes two or three identical immunogenic fragments.

In some embodiments, the recombinant polypeptide includes a plurality of non-identical immunogenic fragments. For example, a recombinant polypeptide can include two, three, four, five, six, seven, eight, nine, ten, or more immunogenic fragments, wherein each fragment comprises, consists of, or consists essentially of a different amino acid sequence from each other fragment, i.e., each fragment is a different fragment. In some embodiments, the recombinant polypeptide includes two, three, four or five different immunogenic fragments. In yet further embodiments, the recombinant polypeptide includes two or three different immunogenic fragments.

In some embodiments, the recombinant polypeptide includes a plurality of immunogenic fragments, in which some of the fragments are identical, but not all. For example, a recombinant polypeptide can include two, three, four, five, six, seven, eight, nine, ten, or more immunogenic fragments, wherein each fragment comprises, consists of, or consists essentially of the same amino acid sequence, while also including one, two, three, four, five, six, seven, eight, nine, ten, or more immunogenic fragments, wherein each fragment comprises, consists of, or consists essentially of a different amino acid sequence from each other fragment. In some embodiments, the recombinant polypeptide includes a total of two, three, four or five immunogenic fragments. In yet further embodiments, the recombinant polypeptide includes a total of two or three immunogenic fragments.

In certain embodiments wherein the recombinant polypeptide comprises an antibody Fc region, the at least one immunogenic fragment can be arranged in any suitable serial orientation with respect to the Fc region. In some embodiments, the at least one immunogenic fragment is connected to the N-terminus of the Fc region. This orientation can be depicted as [immunogenic fragment]_x-[N-terminus-Fc region-C-terminus], wherein X is an integer 1-10 representing the number of immunogenic fragments within the recombinant polypeptide. In some embodiments, the at least one immunogenic fragment is connected to the C-terminus of the Fc region. This orientation can be depicted as [N-terminus-Fc region-C-terminus]-[immunogenic fragment]_x, wherein X is an integer 1-10 representing the number of immunogenic fragments within the recombinant polypeptide. In some embodiments, wherein the recombinant polypeptide includes at least two immunogenic fragments, at least one immunogenic fragment is connected to the N-terminus of the Fc region, and at least one immunogenic fragment is connected to the C-terminus of the Fc region. This orientation can be depicted as [immunogenic fragment]_x-[N-terminus-Fc region-C-terminus]-[immunogenic fragment]_y, wherein X and Y are independently an integer 1-10 representing the number of immunogenic fragments connected to each side of the Fc region. In some embodiments, wherein the recombinant polypeptide includes two immunogenic fragments, both immunogenic

fragments are connected to the N-terminus of the Fc region. This orientation can be depicted as [immunogenic fragment]-[immunogenic fragment]-[N-terminus-Fc region-C-terminus]. In other embodiments, wherein the recombinant polypeptide includes two immunogenic fragments, both immunogenic fragments are connected to the C-terminus of the Fc region. This orientation can be depicted as [N-terminus-Fc region-C-terminus]-[immunogenic fragment]-[immunogenic fragment]. In some embodiments, wherein the recombinant polypeptide includes three immunogenic fragments, each immunogenic fragment is connected to the N-terminus of the Fc region. This orientation can be depicted as [immunogenic fragment]-[immunogenic fragment]-[immunogenic fragment]-[N-terminus-Fc region-C-terminus]. In other embodiments, wherein the recombinant polypeptide includes three immunogenic fragments, each immunogenic fragment is connected to the C-terminus of the Fc region. This orientation can be depicted as [N-terminus-Fc region-C-terminus]-[immunogenic fragment]-[immunogenic fragment]-[immunogenic fragment].

Embodiments of the recombinant polypeptide that include a plurality of immunogenic fragments provide for a flexible expression platform with robust expression of full-length protein, modality to modify individual or multiple domains within one or more of the plurality of immunogenic fragments to reflect the most recent virus variant sequence(s), allow for single step affinity purification by use of an Fc region if present in the recombinant polypeptide, and provide high-level, long-term immune response as tested in Rhesus macaques.

In some embodiments, wherein the recombinant polypeptide includes a plurality of immunogenic fragments, the immunogenic fragments are connected to each other via a linker. The linker can be any suitable linker. Suitable linkers include a polypeptide comprising, consisting of, or consisting essentially of, an amino acid sequence of 1-35 residues, wherein each residue is independently serine, glycine, or aspartic acid, and further wherein the amino acid sequence contains zero or one aspartic acid residues. Other suitable linkers include a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO: 65 (Fc1), SEQ ID NO: 67 (Fc1-TEV), SEQ ID NO: 69 (Fc1-Rv3C), SEQ ID NO: 193 (Short), SEQ ID NO: 195 (Medium), and SEQ ID NO: 197 (Long). When a recombinant polypeptide includes two or more such linkers, the amino acid sequence of the linkers can be identical or different. In some embodiments, wherein the recombinant polypeptide includes a plurality of immunogenic fragments, the immunogenic fragments are connected directly to each other without an intervening linker.

In some embodiments, the recombinant polypeptide comprising or consisting essentially of one or more immunogenic fragments further comprises an antibody Fc region. In some embodiments, the recombinant polypeptide comprising, consisting of, or consisting essentially of one or more immunogenic fragments does not comprise an antibody Fc region. In some embodiments, the recombinant polypeptide comprising or consisting essentially of the one or more immunogenic fragments initially comprises an antibody Fc region, e.g., the recombinant polypeptide comprises an antibody Fc region upon expression of the recombinant polypeptide, wherein the antibody Fc region is subsequently removed, for instance, enzymatically removed. Suitable enzymes include, but are not limited to, class I proteases such as IdeS and IdeZ, and class II enzymes such as the Tobacco Etch Virus (TEV), Rhinovirus 3C (Rv3C), and Factor Xa. When a class II enzyme is employed, the recom-

binant polypeptide comprises the cleavage site for the employed enzyme. In some embodiments, the cleavage site is within or adjacent to the linker region between the Spike protein subunit and the Fc domain.

In some embodiments, removal occurs during a purification step or after the recombinant polypeptide has undergone at least one purification step. For instance, the purification is any suitable type of affinity purification, such as Protein A, Protein G, or Protein M purification. In some embodiments, subtractive chromatography steps are used to remove the enzyme. In subtractive chromatography, unwanted proteins are retained on the resin whilst desired proteins flow-through and are collected for further uses. Examples of suitable subtractive chromatography methods include IMAC for His6-tagged enzymes, and Protein A resin, Protein G resin, or Protein M resin to remove Fc-fusion enzymes. In some embodiments, cut Fcs and uncut Fc-fusions protein are removed by one or more of Protein A, Protein G, and Protein M resin chromatography.

In some embodiments, the one or more immunogenic fragments are connected to the antibody Fc region via a linker. The linker can be any suitable linker. Suitable linkers include a polypeptide comprising an amino acid sequence of 1-35 residues, wherein each residue is independently serine, glycine, or aspartic acid, and further wherein the amino acid sequence contains zero or one aspartic acid residues. Other suitable linkers include a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO: 65 (Fc1), SEQ ID NO: 67 (Fc1-TEV), SEQ ID NO: 69 (Fc1-Rv3C), SEQ ID NO: 193 (Short), SEQ ID NO: 195 (Medium), and SEQ ID NO: 197 (Long). When a recombinant polypeptide includes two or more such linkers, the amino acid sequence of the linkers can be identical or different from each other. In some embodiments, wherein the recombinant polypeptide includes at least one immunogenic fragment connected to the N-terminus of the Fc region, and/or at least one immunogenic fragment is connected to the C-terminus of the Fc region, the immunogenic fragment(s) nearest to the Fc region are connected directly to the Fc region without an intervening linker.

The antibody Fc region included in the recombinant polypeptide can be any suitable antibody Fc region. Suitable antibody Fc regions include wild-type human or other animal Immunoglobulin Fc regions such as IgG, IgA, IgD, IgE, IgM, and their respective subclasses, for example, human IgG1 Fc regions and Fc regions derived therefrom. Other suitable antibody Fc regions include mutant Fc regions that enhance or diminish Fc-receptor binding affinity to speed up or slow down uptake, respectively. In some embodiments, the antibody Fc region comprises the amino acid sequence of SEQ ID NO: 71.

In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]), SEQ ID NO: 75 (LS3472, LS3473, LS3474 [CTD_long_a-Fc]), SEQ ID NO: 77 (LS3477 [CTD_short_a-TEV-Fc]), SEQ ID NO: 79 (LS3485 [CTD_long_a-TEV-Fc]), SEQ ID NO: 81 (LS3489 [CTD_short_a-Rv3c-Fc]), SEQ ID NO: 83 (LS3497 [CTD_long_a-Rv3c-Fc]), SEQ ID NO: 85 (LS2316, LS2317, LS2318, LS2319 [CTD_long_a-His8]), SEQ ID NO: 87 (LS3479 [NTD_short_a-TEV-Fc]), SEQ ID NO: 89 (LS3475 [NTD_long_a-TEV-Fc]), SEQ ID NO: 91 (LS3491 [NTD_short_a-Rv3c-Fc]), SEQ ID NO: 93 (LS3487 [NTD_long_a-Rv3c-Fc]), SEQ ID NO: 95

(LS2326 [NTD_long_a-Fc]), SEQ ID NO: 97 (LS2354 [CTD_long_a-D614G-Fc]), SEQ ID NO: 99 (LS2355 [CTD_long_a-Dimer-Fc]), SEQ ID NO: 101 (LS2356 [CTD_long_b-Fc]), SEQ ID NO: 103 (LS2357 [CTD_long_c-Fc]), SEQ ID NO: 105 (LS2358 [CTD_long_d-Fc]), SEQ ID NO: 107 (LS2359 [CTD_long_e-Fc]), SEQ ID NO: 109 (LS2360 [CTD_long_f-Fc]), SEQ ID NO: 111 (LS2361 [CTD_long_g-Fc]), SEQ ID NO: 113 (LS2362 [CTD_long_h-Fc]), SEQ ID NO: 115 (LS2363 [CTD_long_i-Fc]), SEQ ID NO: 117 (LS2364 [CTD_short_b-Fc]), SEQ ID NO: 119 (LS2365 [CTD_short_c-Fc]), SEQ ID NO: 121 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 123 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 125 (LS2368 [CTD_short_f-Fc]), SEQ ID NO: 127 (LS2369 [CTD_short_g-Fc]), SEQ ID NO: 129 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 131 (LS2371 [CTD_short_i-Fc]), SEQ ID NO: 133 (LS2372 [CTD_vs_a-Fc]), SEQ ID NO: 135 (LS2373 [CTD_vs_b-Fc]), SEQ ID NO: 137 (LS2374 [CTD_vs_c-Fc]), SEQ ID NO: 139 (LS2375 [CTD_vs_d-Fc]), SEQ ID NO: 141 (LS2376 [CTD_vs_e-Fc]), SEQ ID NO: 143 (LS2377 [RBD_a-Fc]), SEQ ID NO: 145 (LS2378 [RBD_b-Fc]), SEQ ID NO: 147 (LS2379 [RBD_c-Fc]), SEQ ID NO: 149 (LS2380 [RBD_d-Fc]), SEQ ID NO: 151 (LS2381 [RBD_e-Fc]), SEQ ID NO: 153 (LS2382 [NTD_short_a-Fc]), SEQ ID NO: 155 (LS2393 [(RBD-tight)₂-Fc]), SEQ ID NO: 157 (LS2394 [(RBD-extended)₂-Fc]), SEQ ID NO: 159 (LS2395 [(RBD)₂-Fc]), SEQ ID NO: 161 (LS2397-2400 [(CTD_short_d)₂-Fc]), SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i)₂-Fc]), SEQ ID NO: 165 (LS2421 and LS2422 [(CTD_short_i)₂-Fc]-mod. 1], SEQ ID NO: 167 (LS2423 [(CTD_short_i)₂-Fc]-mod. 2], SEQ ID NO: 169 (LS2435 [(CTD_short_i)₂-Fc]-mod. 3], SEQ ID NO: 217, SEQ ID NO: 219, SEQ ID NO: 221, SEQ ID NO: 223, SEQ ID NO: 225, SEQ ID NO: 227, SEQ ID NO: 229, SEQ ID NO: 231, SEQ ID NO: 233, SEQ ID NO: 235, SEQ ID NO: 237, SEQ ID NO: 237, SEQ ID NO: 239, SEQ ID NO: 241, SEQ ID NO: 243, SEQ ID NOs: 255-264 (BA.X-1-monomer-Fc-BA.X-10-monomer-Fc), and SEQ ID NOs: 275-284 (BA.X-1-dimer-Fc-BA.X-10-dimer-Fc). In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i)₂-Fc]), SEQ ID NO: 165 (LS2421 and LS2422 [(CTD_short_i)₂-Fc]-mod. 1], SEQ ID NO: 167 (LS2423 [(CTD_short_i)₂-Fc]-mod. 2], and SEQ ID NO: 169 (LS2435 [(CTD_short_i)₂-Fc]-mod. 3]. In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence of SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i)₂-Fc]). In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 161 (LS2397-2400 [(CTD_short_d)₂-Fc]), SEQ ID NO: 121 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 123 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 129 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 131 (LS2371 [CTD_short_i-Fc]), and SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]). In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid

LS2422 [(CTD_short_i)₂-Fc], SEQ ID NO: 167 (LS2423 [(CTD_short_i)₂-Fc], and SEQ ID NO: 169 (LS2435 [(CTD_short_i)₂-Fc]). In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, the amino acid sequence of SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i)₂-Fc]). In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence selected from the group consisting of SEQ ID NO: 161 (LS2397-2400 [(CTD_short_d)₂-Fc]), SEQ ID NO: 121 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 123 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 129 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 131 (LS2371 [CTD_short_i-Fc]), and SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]). In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, the amino acid sequence represented by SEQ ID NO: 163. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, the amino acid sequence represented by SEQ ID NO: 169. In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence selected from the group consisting of SEQ ID NOs: 255-264 and SEQ ID NOs: 275-284. In further embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence selected from the group consisting of SEQ ID NOs: 255-264. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 255. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 256. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 257. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 258. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 259. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 260. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 261. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 262. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 263. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 264. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 275. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 276. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 277. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 278. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of,

an amino acid sequence represented by SEQ ID NO: 279. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 280. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 281. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 282. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 283. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 284.

In some embodiments, the recombinant polypeptide is encoded by a recombinant polynucleotide. In certain embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to a nucleic acid sequence selected from the group consisting of SEQ ID NO: 74 (LS2330 [CTD_short_a-Fc]), SEQ ID NO: 76 (LS3472, LS3473, LS3474 [CTD_long_a-Fc]), SEQ ID NO: 78 (LS3477 [CTD_short_a-TEV-Fc]), SEQ ID NO: 80 (LS3485 [CTD_long_a-TEV-Fc]), SEQ ID NO: 82 (LS3489 [CTD_short_a-Rv3c-Fc]), SEQ ID NO: 84 (LS3497 [CTD_long_a-Rv3c-Fc]), SEQ ID NO: 86 (LS2316, LS2317, LS2318, LS2319 [CTD_long_a-His8]), SEQ ID NO: 88 (LS3479 [NTD_short_a-TEV-Fc]), SEQ ID NO: 90 (LS3475 [NTD_long_a-TEV-Fc]), SEQ ID NO: 92 (LS3491 [NTD_short_a-Rv3c-Fc]), SEQ ID NO: 94 (LS3487 [NTD_long_a-Rv3c-Fc]), SEQ ID NO: 96 (LS2326 [NTD_long_a-Fc]), SEQ ID NO: 98 (LS2354 [CTD_long_a-D614G-Fc]), SEQ ID NO: 100 (LS2355 [CTD_long_a-Dimer-Fc]), SEQ ID NO: 102 (LS2356 [CTD_long_b-Fc]), SEQ ID NO: 104 (LS2357 [CTD_long_c-Fc]), SEQ ID NO: 106 (LS2358 [CTD_long_d-Fc]), SEQ ID NO: 108 (LS2359 [CTD_long_e-Fc]), SEQ ID NO: 110 (LS2360 [CTD_long_f-Fc]), SEQ ID NO: 112 (LS2361 [CTD_long_g-Fc]), SEQ ID NO: 114 (LS2362 [CTD_long_h-Fc]), SEQ ID NO: 116 (LS2363 [CTD_long_i-Fc]), SEQ ID NO: 118 (LS2364 [CTD_short_b-Fc]), SEQ ID NO: 120 (LS2365 [CTD_short_c-Fc]), SEQ ID NO: 122 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 124 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 126 (LS2368 [CTD_short_f-Fc]), SEQ ID NO: 128 (LS2369 [CTD_short_g-Fc]), SEQ ID NO: 130 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 132 (LS2371 [CTD_short_i-Fc]), SEQ ID NO: 134 (LS2372 [CTD_vs_a-Fc]), SEQ ID NO: 136 (LS2373 [CTD_vs_b-Fc]), SEQ ID NO: 138 (LS2374 [CTD_vs_c-Fc]), SEQ ID NO: 140 (LS2375 [CTD_vs_d-Fc]), SEQ ID NO: 142 (LS2376 [CTD_vs_e-Fc]), SEQ ID NO: 144 (LS2377 [RBD_a-Fc]), SEQ ID NO: 146 (LS2378 [RBD_b-Fc]), SEQ ID NO: 148 (LS2379 [RBD_c-Fc]), SEQ ID NO: 150 (LS2380 [RBD_d-Fc]), SEQ ID NO: 152 (LS2381 [RBD_e-Fc]), SEQ ID NO: 154 (LS2382 [NTD_short_a-Fc]), SEQ ID NO: 156 (LS2393 [(RBD-tight)₂-Fc]), SEQ ID NO: 158 (LS2394 [(RBD-extended)₂-Fc]), SEQ ID NO: 160 (LS2395 [(RBD)₂-Fc]), SEQ ID NO: 162 (LS2397-2400 [(CTD_short_d)₂-Fc]), SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i)₂-Fc]), SEQ ID NO: 166 (LS2421 and LS2422 [(CTD_short_i)₂-Fc]), SEQ ID NO: 168 (LS2423 [(CTD_short_i)₂-Fc]), SEQ ID NO: 170 (LS2435 [(CTD_short_i)₂-Fc]), SEQ ID NO: 218, SEQ ID NO: 220, SEQ ID NO: 222, SEQ ID NO: 224, SEQ ID NO: 226, SEQ

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ID NO: 228, SEQ ID NO: 230, SEQ ID NO: 232, SEQ ID NO: 234, SEQ ID NO: 236, SEQ ID NO: 238, SEQ ID NO: 240, SEQ ID NO: 242, SEQ ID NO: 244, SEQ ID NOs: 295-304, and SEQ ID NOs: 315-324.

In some embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to a nucleic acid sequence selected from the group consisting of SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i)₂-Fc]), SEQ ID NO: 166 (LS2421 and LS2422 [(CTD_short_i)₂-Fc]-mod.1), SEQ ID NO: 168 (LS2423 [(CTD_short_i)₂-Fc]-mod. 2]), and SEQ ID NO: 170 (LS2435 [(CTD_short_i)₂-Fc-mod. 3]). In some embodiments, the polynucleotide comprises, consists of, or consists essentially of, an nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the nucleic acid sequence of SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i)₂-Fc]). In some embodiments, the polynucleotide comprises, consists of, or consists essentially of, an nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an nucleic acid sequence selected from the group consisting of SEQ ID NO: 162 (LS2397-2400 [(CTD_short_d)₂-Fc]), SEQ ID NO: 122 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 124 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 130 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 132 (LS2371 [CTD_short_i-Fc]), and SEQ ID NO: 74 (LS2330 [CTD_short_a-Fc]). In an embodiment, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the nucleic acid sequence represented by SEQ ID NO: 164. In an embodiment, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 95%, 96%, 97%, 98%, or 99%, sequence identity to the nucleic acid sequence represented by SEQ ID NO: 170. In certain embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the nucleic acid sequence selected from the group consisting of SEQ ID NOs: 295-304 and SEQ ID NOs: 315-324. In certain embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the nucleic acid sequence selected from the group consisting of SEQ ID NOs: 295-304.

In some embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence selected from the group consisting of SEQ ID NO: 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240, 242, 244, 295-304, and 315-324. In some embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence selected from the group consisting of SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i)₂-Fc]), SEQ ID NO: 166 (LS2421 and LS2422 [(CTD_short_i)₂-Fc]-mod. 1), SEQ ID NO: 168 (LS2423 [(CTD_short_i)₂-Fc]-mod.2]), and SEQ ID NO: 170 (LS2435 [(CTD_short_i)₂-Fc]-mod. 3]). In some embodiments, the polynucleotide comprises, consists of, or consists essentially of, the nucleic acid

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sequence of SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i)₂-Fc]). In some embodiments, polynucleotide comprises, consists of, or consists essentially of, an nucleic acid sequence selected from the group consisting of SEQ ID NO: 162 (LS2397-2400 [(CTD_short_d)₂-Fc]), SEQ ID NO: 122 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 124 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 130 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 132 (LS2371 [CTD_short_i-Fc]), and SEQ ID NO: 74 (LS2330 [CTD_short_a-Fc]). In an embodiment, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence represented by SEQ ID NO: 164. In an embodiment, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence represented by SEQ ID NO: 170. In certain embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 295-304 and SEQ ID NOs: 315-324. In certain embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 295-304.

Nucleic acid or amino acid sequence "identity," as referenced herein, can be determined by comparing a nucleic acid or amino acid sequence of interest to a reference nucleic acid or amino acid sequence. The percent identity is the number of nucleotides or amino acid residues that are the same (i.e., that are identical) as between the optimally aligned sequence of interest and the reference sequence divided by the length of the longest sequence (i.e., the length of either the sequence of interest or the reference sequence, whichever is longer). Alignment of sequences and calculation of percent identity can be performed using available software programs. Examples of such programs include CLUSTAL-W, T-Coffee, and ALIGN (for alignment of nucleic acid and amino acid sequences), BLAST programs (e.g., BLAST 2.1, BL2SEQ, BLASTp, BLASTn, and the like) and FASTA programs (e.g., FASTA3x, FASTM, and SSEARCH) (for sequence alignment and sequence similarity searches). Sequence alignment algorithms also are disclosed in, for example, Altschul et al., *J. Molecular Biol.*, 215(3): 403-410 (1990), Beigert et al., *Proc. Natl. Acad. Sci. USA*, 106(10): 3770-3775 (2009), Durbin et al., eds., *Biological Sequence Analysis: Probabilistic Models of Proteins and Nucleic Acids*, Cambridge University Press, Cambridge, UK (2009), Soding, *Bioinformatics*, 21(7): 951-960 (2005), Altschul et al., *Nucleic Acids Res.*, 25(17): 3389-3402 (1997), and Gusfield, *Algorithms on Strings, Trees and Sequences*, Cambridge University Press, Cambridge UK (1997)). Percent (%) identity of sequences can be also calculated, for example, as $100 \times [(\text{identical positions}) / \min(\text{TG}_A, \text{TG}_B)]$, where TG_A and TG_B are the sum of the number of residues and internal gap positions in peptide sequences A and B in the alignment that minimizes TG_A and TG_B . See, e.g., Russell et al., *J. Mol. Biol.*, 244: 332-350 (1994).

In some embodiments, the nucleotide sequence of the polynucleotide is codon optimized and/or codon pair optimized.

Another embodiment is a recombinant vector that comprises, consists of, or consists essentially of, a polynucleotide that encodes the recombinant polypeptide described herein. The recombinant vector can be any suitable vector. Examples of suitable recombinant vectors include but are not limited to a pcDNA3.1, a pSV, a pCMV, a pBapo-CMV, or a pBapo-EF1alpha expression vector.

Yet another embodiment is an isolated cell that includes the recombinant polypeptide described herein or a recom-

binant polynucleotide that contains a nucleic acid sequence that encodes the recombinant polypeptide.

Another embodiment is a pharmaceutical composition that contains the recombinant polypeptide described herein and at least one pharmaceutically acceptable carrier. The at least one pharmaceutically acceptable carrier can be any suitable carrier. Examples of suitable carriers include water and any suitable buffer. Suitable buffers include HEPES-buffered saline and phosphate-buffered saline. In certain embodiments, the pharmaceutical composition further contains at least one adjuvant. The at least one adjuvant can be any suitable adjuvant. Examples of suitable adjuvants include alum adjuvants, emulsion adjuvants, and pattern recognition receptor agonist adjuvants. Further examples include AS03, MF59, Squalene Emulsion, Alum, aluminum hydroxide gels, calcium phosphate hydroxide, paraffin oil, cytokines (IL-1, IL-2, IL-12), killed bacterial products such as *Bordetella* and *Mycobacterium* bacteria, bacterial toxoids, squalene and DL- α -tocopherol emulsions, squalene-oil-in-water emulsion, aluminum phosphate gels, saponins, cyclic dinucleotides, and TLR agonists, preferably TLR1, TLR2, TLR4, TLR5, TLR7, TLR8, TLR9 etc., and combinations thereof. In certain embodiments, the adjuvant is AS03. In certain other embodiments, the adjuvant is Alum. In yet further embodiments, the pharmaceutical composition does not contain an adjuvant.

The pharmaceutical composition can contain any therapeutically effective amount of the recombinant polypeptide described herein. A therapeutically effective amount is an amount sufficient to induce an immune response against the target virus or viruses, for instance, SARS-CoV-2, SARS-CoV, and/or MERS-CoV, preferably SARS-CoV-2. Typically, a dosage is therapeutically effective if it prevents, inhibits, reduces, eliminates, protects against, and/or delays the onset of an infection or an infectious clinical condition caused by a beta coronavirus in a subject. Infectious clinical conditions include, for example, fever or chills, cough, shortness of breath or difficult breathing, fatigue, muscle or body aches, headache, loss of taste or smell, sore throat, congestion, runny nose, nausea, vomiting, and diarrhea. In some embodiments, a single dose of the pharmaceutical composition contains 10 nanograms to 1 milligram, 0.1-250 micrograms, 10-100 micrograms, or 12.5-50 micrograms of the recombinant polypeptide. In some embodiments, a single dose of the pharmaceutical composition contains 0.01-1, 0.1-1, 0.5-5, 1-20, or 5-15, 1-50, or 10-50 micrograms of the recombinant polypeptide. In some embodiments, wherein the pharmaceutical composition does not contain an adjuvant, a single dose of the pharmaceutical composition could contain 0.1-1, 0.1-1, 0.5-5, 1-20, or 5-15, 1-50, or 10-50 micrograms of the recombinant polypeptide. However, a single dose of the pharmaceutical composition could also contain increased amounts of the recombinant polypeptide, such as 100-1000, 100-250, or 250-500 micrograms of recombinant polypeptide.

A further embodiment is a method for preventing, inhibiting, reducing, eliminating, protecting against, or delaying the onset of an infection or an infectious clinical condition caused by a beta coronavirus in a subject comprising administering to the subject the recombinant polypeptide described herein, the polypeptide encoded by the recombinant polynucleotide described herein, or a dose of the pharmaceutical composition described herein. Examples of beta coronaviruses include SARS-CoV, MERS-CoV, and SARS-CoV-2.

Another embodiment is a method for inducing an immune response against a beta coronavirus in a subject comprising administering to the subject the recombinant polypeptide

described herein, the polypeptide encoded by the recombinant polynucleotide described herein, or a dose of the pharmaceutical composition described herein. Examples of beta coronaviruses include SARS-CoV, MERS-CoV, and SARS-CoV-2.

In the methods described herein, administration can be achieved by any suitable administration method. Suitable administration methods include oral, parenteral, subcutaneous, intravenous, intramuscular, intrapulmonary, intranasal, intraarterial, intrathecal, and intraperitoneal administration.

In the methods described herein, the subject can be any suitable animal that is capable of being infected by a beta coronavirus, such as SARS-CoV, MERS-CoV, and SARS-CoV-2. The subject can be a human, non-human primate, horse, pig, cattle, cat, dog, sheep, mink, rodent, hamster, or bat, preferably human.

Examples of Non-Limiting Aspects of the Disclosure

Aspects, including embodiments, of the invention described herein may be beneficial alone or in combination, with one or more other aspects or embodiments. Without limiting the foregoing description, certain non-limiting aspects of the disclosure numbered (1)-(36) are provided below. As will be apparent to those of skill in the art upon reading this disclosure, each of the individually numbered aspects may be used or combined with any of the preceding or following individually numbered aspects. This is intended to provide support for all such combinations of aspects and is not limited to combinations of aspects explicitly provided below:

(1) A recombinant polypeptide comprising at least one immunogenic fragment of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein and optionally an antibody Fc region.

(2) The recombinant polypeptide of aspect 1, wherein the at least one fragment of the SARS-CoV-2 spike glycoprotein comprises the N-terminal domain of the S1 subunit, the C-terminal domain of the S1 subunit, or both.

(3) The recombinant polypeptide of aspect 1 or 2, wherein the at least one fragment of the SARS-CoV-2 spike glycoprotein comprises an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 1 (CTD_long_a), SEQ ID NO: 3 (CTD_long_a D614G), SEQ ID NO: 5 (CTD_long_a-Dimer), SEQ ID NO: 7 (CTD_long_b), SEQ ID NO: 9 (CTD_long_c), SEQ ID NO: 11 (CTD_long_d), SEQ ID NO: 13 (CTD_long_e), SEQ ID NO: 15 (CTD_long_f), SEQ ID NO: 17 (CTD_long_g), SEQ ID NO: 19 (CTD_long_h), SEQ ID NO: 21 (CTD_long_i), SEQ ID NO: 23 (CTD_short_a), SEQ ID NO: 25 (CTD_short_b), SEQ ID NO: 27 (CTD_short_c), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 33 (CTD_short_f), SEQ ID NO: 35 (CTD_short_g), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 41 (CTD_vs_a), SEQ ID NO: 43 (CTD_vs_b), SEQ ID NO: 45 (CTD_vs_c), SEQ ID NO: 47 (CTD_vs_d), SEQ ID NO: 49 (CTD_vs_e), SEQ ID NO: 51 (RBD_a), SEQ ID NO: 53 (RBD_b), SEQ ID NO: 55 (RBD_c), SEQ ID NO: 57 (RBD_d), SEQ ID NO: 59 (RBD_e), SEQ ID NO: 61 (NTD_long_a), SEQ ID NO: 63 (NTD_short_a), SEQ ID NO: 171 (RBD-tight), SEQ ID NO: 173 ((RBD-tight)₂), SEQ ID NO: 175 (RBD-extended), SEQ ID NO: 177 ((RBD-extended)₂), SEQ ID NO: 179 (RBD), SEQ ID NO:

181 ((RBD)₂), SEQ ID NO: 183 ((CTD_short_d)₂), SEQ ID NO: 185 ((CTD_short_i)₂), SEQ ID NO: 187 ((CTD_short_i)₂—mod. 1), SEQ ID NO: 189 ((CTD_short_i)₂—mod. 2), SEQ ID NO: 191 ((CTD_short_i)₂—mod. 3), SEQ ID NOS: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOS: 265-274 (BA.X-1-dimer-BA.X-10-dimer).

(4) The recombinant polypeptide of any one of aspects 1-3, wherein the polypeptide comprises an amino acid sequence selected from the group consisting of SEQ ID NOS: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 171, 173, 175, 177, 179, 181, 183, 185, 187, 189, 191, 199, 201, 203, 205, 207, 209, 211, 213, 215, 245-254, and 265-274.

(5) The recombinant polypeptide of any of one aspects 1-4, wherein the polypeptide comprises at least two immunogenic fragments.

(6) The recombinant polypeptide of aspect 5, wherein each of the at least two immunogenic fragments comprises an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 1 (CTD_long_a), SEQ ID NO: 3 (CTD_long_a D614G), SEQ ID NO: 5 (CTD_long_a-Dimer), SEQ ID NO: 7 (CTD_long_b), SEQ ID NO: 9 (CTD_long_c), SEQ ID NO: 11 (CTD_long_d), SEQ ID NO: 13 (CTD_long_e), SEQ ID NO: 15 (CTD_long_f), SEQ ID NO: 17 (CTD_long_g), SEQ ID NO: 19 (CTD_long_h), SEQ ID NO: 21 (CTD_long_i), SEQ ID NO: 23 (CTD_short_a), SEQ ID NO: 25 (CTD_short_b), SEQ ID NO: 27 (CTD_short_c), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 33 (CTD_short_f), SEQ ID NO: 35 (CTD_short_g), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 41 (CTD_vs_a), SEQ ID NO: 43 (CTD_vs_b), SEQ ID NO: 45 (CTD_vs_c), SEQ ID NO: 47 (CTD_vs_d), SEQ ID NO: 49 (CTD_vs_e), SEQ ID NO: 51 (RBD_a), SEQ ID NO: 53 (RBD_b), SEQ ID NO: 55 (RBD_c), SEQ ID NO: 57 (RBD_d), SEQ ID NO: 59 (RBD_e), SEQ ID NO: 61 (NTD_long_a), SEQ ID NO: 63 (NTD_short_a), SEQ ID NO: 171 (RBD-tight), SEQ ID NO: 173 ((RBD-tight)₂), SEQ ID NO: 175 (RBD-extended), SEQ ID NO: 177 ((RBD-extended)₂), SEQ ID NO: 179 (RBD), SEQ ID NO: 181 ((RBD)₂), SEQ ID NO: 183 ((CTD_short_d)₂), SEQ ID NO: 185 ((CTD_short_i)₂), SEQ ID NO: 187 ((CTD_short_i)₂—mod. 1), SEQ ID NO: 189 ((CTD_short_i)₂—mod. 2), SEQ ID NO: 191 ((CTD_short_i)₂—mod. 3), SEQ ID NO: 199 (SARS-2003, SARS_short_h), SEQ ID NO: 201 (SARS-2003, SARS_short_i), SEQ ID NO: 203 (MERS_Lytic_a), SEQ ID NO: 205 (MERS_Lytic_b), SEQ ID NO: 207 (MERS_Lytic_c), SEQ ID NO: 209 (MERS_Lytic_d), SEQ ID NO: 211 (MERS_Lytic_e), SEQ ID NO: 213 (MERS_Lytic_f), SEQ ID NO: 215 (MERS_Lytic_g), SEQ ID NOS: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOS: 265-274 (BA.X-1-dimer-BA.X-10-dimer).

(7) The recombinant polypeptide of aspect 5, wherein each of the at least two immunogenic fragments comprises an amino acid sequence selected from the group consisting of SEQ ID NOS: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 171, 173, 175, 177, 179, 181, 183, 185, 187, 189, 191, 199, 201, 203, 205, 207, 209, 211, 213, 215, 245-254, and 265-274.

(8) The recombinant polypeptide of any one of aspects 5-7, wherein each immunogenic fragment of the at least two immunogenic fragments comprises the same amino acid sequence.

(9) The recombinant polypeptide of any one of aspects 5-8, wherein each immunogenic fragment of the at least two immunogenic fragments comprises a different amino acid sequence from the other immunogenic fragments.

(10) The recombinant polypeptide of any one of aspects 1-9, wherein the polypeptide comprises two, three, four, or five immunogenic fragments.

(11) The recombinant polypeptide of any one of aspects 5-10, wherein the at least two immunogenic fragments are connected to each other via a linker.

(12) The recombinant polypeptide of aspect 11, wherein the linker is a polypeptide comprising an amino acid sequence of 1-35 residues, wherein each residue is independently serine or glycine.

(13) The recombinant polypeptide of any one of aspects 1-12, wherein the at least one immunogenic fragment of the SARS-CoV-2 spike glycoprotein is connected to the antibody Fc region via a linker.

(14) The recombinant polypeptide of aspect 13, wherein the linker comprises an amino acid sequence selected from the group consisting of SEQ ID NO: 65 (Fc1), SEQ ID NO: 67 (Fc1-TEV), SEQ ID NO: 69 (Fc1-Rv3C), SEQ ID NO: 193 (Short), SEQ ID NO: 195 (Medium), and SEQ ID NO: 197 (Long).

(15) The recombinant polypeptide of any one of aspects 1-14, wherein the antibody Fc region is from a human IgG1 antibody or derived therefrom.

(16) The recombinant polypeptide of aspect 15, wherein the antibody Fc region comprises the amino acid sequence of SEQ ID NO: 71.

(17) The recombinant polypeptide of any one of aspects 1-16, wherein the polypeptide comprises an amino acid sequence with at least 90%, preferably at least 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]), SEQ ID NO: 75 (LS3472, LS3473, LS3474 [CTD_long_a-Fc]), SEQ ID NO: 77 (LS3477 [CTD_short_a-TEV-Fc]), SEQ ID NO: 79 (LS3485 [CTD_long_a-TEV-Fc]), SEQ ID NO: 81 (LS3489 [CTD_short_a-Rv3c-Fc]), SEQ ID NO: 83 (LS3497 [CTD_long_a-Rv3c-Fc]), SEQ ID NO: 85 (LS2316, LS2317, LS2318, LS2319 [CTD_long_a-Hiss]), SEQ ID NO: 87 (LS3479 [NTD_short_a-TEV-Fc]), SEQ ID NO: 89 (LS3475 [NTD_long_a-TEV-Fc]), SEQ ID NO: 91 (LS3491 [NTD_short_a-Rv3c-Fc]), SEQ ID NO: 93 (LS3487 [NTD_long_a-Rv3c-Fc]), SEQ ID NO: 95 (LS2326 [NTD_long_a-Fc]), SEQ ID NO: 97 (LS2354 [CTD_long_a D614G-Fc]), SEQ ID NO: 99 (LS2355 [CTD_long_a-Dimer-Fc]), SEQ ID NO: 101 (LS2356 [CTD_long_b-Fc]), SEQ ID NO: 103 (LS2357 [CTD_long_c-Fc]), SEQ ID NO: 105 (LS2358 [CTD_long_d-Fc]), SEQ ID NO: 107 (LS2359 [CTD_long_e-Fc]), SEQ ID NO: 109 (LS2360 [CTD_long_f-Fc]), SEQ ID NO: 111 (LS2361 [CTD_long_g-Fc]), SEQ ID NO: 113 (LS2362 [CTD_long_h-Fc]), SEQ ID NO: 115 (LS2363 [CTD_long_i-Fc]), SEQ ID NO: 117 (LS2364 [CTD_short_b-Fc]), SEQ ID NO: 119 (LS2365 [CTD_short_c-Fc]), SEQ ID NO: 121 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 123 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 125 (LS2368 [CTD_short_f-Fc]), SEQ ID NO: 127 (LS2369 [CTD_short_g-Fc]), SEQ ID NO: 129 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 131 (LS2371

[CTD_short_i-Fc]), SEQ ID NO: 133 (LS2372 [CTD_vs_a-Fc]), SEQ ID NO: 135 (LS2373 [CTD_vs_b-Fc]), SEQ ID NO: 137 (LS2374 [CTD_vs_c-Fc]), SEQ ID NO: 139 (LS2375 [CTD_vs_d-Fc]), SEQ ID NO: 141 (LS2376 [CTD_vs_e-Fc]), SEQ ID NO: 143 (LS2377 [RBD_a-Fc]), SEQ ID NO: 145 (LS2378 [RBD_b-Fc]), SEQ ID NO: 147 (LS2379 [RBD_c-Fc]), SEQ ID NO: 149 (LS2380 [RBD_d-Fc]), SEQ ID NO: 151 (LS2381 [RBD_e-Fc]), SEQ ID NO: 153 (LS2382 [NTD_short_a-Fc]), SEQ ID NO: 155 (LS2393 [(RBD-tight)₂-Fc]), SEQ ID NO: 157 (LS2394 [(RBD-extended)₂-Fc]), SEQ ID NO: 159 (LS2395 [(RBD)₂-Fc]), SEQ ID NO: 161 (LS2397-2400 [(CTD_short_d)₂-Fc]), SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i)₂-Fc]), SEQ ID NO: 165 (LS2421 and LS2422 [(CTD_short_i)₂-Fc]-mod. 1), SEQ ID NO: 167 (LS2423 [(CTD_short_i)₂-Fc]-mod. 2), SEQ ID NO: 169 (LS2435 [(CTD_short_i)₂-Fc]-mod. 3), SEQ ID NOS: 255-264, and SEQ ID NOS: 275-284.

(18) The recombinant polypeptide of aspect 17, wherein the polypeptide comprises an amino acid sequence selected from the group consisting of SEQ ID NOS: 73, 75, 77, 79, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, 119, 121, 123, 125, 127, 129, 131, 133, 135, 137, 139, 141, 143, 145, 147, 149, 151, 153, 155, 157, 159, 161, 163, 165, 167, 169, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 239, 241, 243, 255-264, and 275-284.

(19) A recombinant polynucleotide encoding the recombinant polypeptide of any one of aspects 1-18.

(20) The recombinant polynucleotide of aspect 19, wherein the polynucleotide comprises a nucleic acid sequence with at least 90%, preferably at least 95%, 96%, 97%, 98%, or 99%, sequence identity to a nucleic acid sequence selected from the group consisting of SEQ ID NO: 74 (LS2330 [CTD_short_a-Fc]), SEQ ID NO: 76 (LS3472, LS3473, LS3474 [CTD_long_a-Fc]), SEQ ID NO: 78 (LS3477 [CTD_short_a-TEV-Fc]), SEQ ID NO: 80 (LS3485 [CTD_long_a-TEV-Fc]), SEQ ID NO: 82 (LS3489 [CTD_short_a-Rv3c-Fc]), SEQ ID NO: 84 (LS3497 [CTD_long_a-Rv3c-Fc]), SEQ ID NO: 86 (LS2316, LS2317, LS2318, LS2319 [CTD_long_a-Hiss]), SEQ ID NO: 88 (LS3479 [NTD_short_a-TEV-Fc]), SEQ ID NO: 90 (LS3475 [NTD_long_a-TEV-Fc]), SEQ ID NO: 92 (LS3491 [NTD_short_a-Rv3c-Fc]), SEQ ID NO: 94 (LS3487 [NTD_long_a-Rv3c-Fc]), SEQ ID NO: 96 (LS2326 [NTD_long_a-Fc]), SEQ ID NO: 98 (LS2354 [CTD_long_a D614G-Fc]), SEQ ID NO: 100 (LS2355 [CTD_long_a-Dimer-Fc]), SEQ ID NO: 102 (LS2356 [CTD_long_b-Fc]), SEQ ID NO: 104 (LS2357 [CTD_long_c-Fc]), SEQ ID NO: 106 (LS2358 [CTD_long_d-Fc]), SEQ ID NO: 108 (LS2359 [CTD_long_e-Fc]), SEQ ID NO: 110 (LS2360 [CTD_long_f-Fc]), SEQ ID NO: 112 (LS2361 [CTD_long_g-Fc]), SEQ ID NO: 114 (LS2362 [CTD_long_h-Fc]), SEQ ID NO: 116 (LS2363 [CTD_long_i-Fc]), SEQ ID NO: 118 (LS2364 [CTD_short_b-Fc]), SEQ ID NO: 120 (LS2365 [CTD_short_c-Fc]), SEQ ID NO: 122 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 124 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 126 (LS2368 [CTD_short_f-Fc]), SEQ ID NO: 128 (LS2369 [CTD_short_g-Fc]), SEQ ID NO: 130 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 132 (LS2371 [CTD_short_i-Fc]), SEQ ID NO: 134 (LS2372 [CTD_vs_a-Fc]), SEQ ID NO: 136 (LS2373 [CTD_vs_b-Fc]), SEQ ID NO: 138 (LS2374 [CTD_vs_c-Fc]), SEQ ID NO: 140 (LS2375 [CTD_vs_d-Fc]), SEQ ID NO: 142 (LS2376

[CTD_vs_e-Fc]), SEQ ID NO: 144 (LS2377 [RBD_a-Fc]), SEQ ID NO: 146 (LS2378 [RBD_b-Fc]), SEQ ID NO: 148 (LS2379 [RBD_c-Fc]), SEQ ID NO: 150 (LS2380 [RBD_d-Fc]), SEQ ID NO: 152 (LS2381 [RBD_e-Fc]), SEQ ID NO: 154 (LS2382 [NTD_short_a-Fc]), SEQ ID NO: 156 (LS2393 [(RBD-tight)₂-Fc]), SEQ ID NO: 158 (LS2394 [(RBD-extended)₂-Fc]), SEQ ID NO: 160 (LS2395 [(RBD)₂-Fc]), SEQ ID NO: 162 (LS2397-2400 [(CTD_short_d)₂-Fc]), SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i)₂-Fc]), SEQ ID NO: 166 (LS2421 and LS2422 [(CTD_short_i)₂-Fc]-mod. 1), SEQ ID NO: 168 (LS2423 [(CTD_short_i)₂-Fc]-mod. 1), and SEQ ID NO: 170 (LS2435 [(CTD_short_i)₂-Fc]-mod. 3), SEQ ID NOS: 295-304, and SEQ ID NOS: 315-324.

(21) The recombinant polynucleotide of aspect 20, wherein the polynucleotide comprises a nucleic acid sequence selected from the group consisting of SEQ ID NO: 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240, 242, 244, 295-304, and 315-324.

(22) The recombinant polynucleotide of any one of aspects 19-21, wherein the nucleic acid sequence has been codon optimized.

(23) A pharmaceutical composition comprising the recombinant polypeptide of any one of aspects 1-18 or the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, and a pharmaceutically acceptable carrier.

(24) The pharmaceutical composition of aspect 23, wherein the pharmaceutical composition further comprises at least one adjuvant.

(25) The pharmaceutical composition of aspect 24, wherein the at least one adjuvant is selected from the group consisting of alum adjuvants, emulsion adjuvants, and pattern recognition receptor agonist adjuvants.

(26) The pharmaceutical composition of aspect 25, wherein the at least one adjuvant is MF59, Squalene Emulsion, Alum, aluminum hydroxide gels, calcium phosphate hydroxide, paraffin oil, cytokines (IL-1, IL-2, IL-12), killed bacterial products such as *Bordetella* and *Mycobacterium* bacteria, bacterial toxoids, squalene and DL- α -tocopherol emulsions, aluminum phosphate gels, saponins, cyclic dinucleotides, and TLR agonists, preferably TLR1, TLR2, TLR4, TLR5, TLR7, TLR8, TLR9 etc., and combinations thereof.

(27) The pharmaceutical composition of aspect 24, wherein the at least one adjuvant is a squalene-oil-in-water emulsion adjuvant.

(28) A vector comprising the recombinant polynucleotide of any one of aspects 19-22.

(29) An isolated cell comprising the recombinant polypeptide of any one of aspects 1-18, or the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22.

(30) A method for preventing, inhibiting, reducing, eliminating, protecting, or delaying the onset of an infection or an infectious clinical condition caused by a beta coronavirus in a subject comprising administering to the subject the recombinant polypeptide of any one of aspects 1-18, the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, or the pharmaceutical composition of any one of aspects 23-27.

(31) A method for inducing an immune response against a beta coronavirus in a subject comprising administering to the subject the recombinant polypeptide of any one of aspects 1-18, the polypeptide encoded by the recombinant

polynucleotide of any one of aspects 19-22, or the pharmaceutical composition of any one of aspects 23-27.

(32) The method of aspect 30 or 31, wherein the recombinant polypeptide of any one of aspects 1-18, the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, or the pharmaceutical composition of any one of aspects 23-27 is administered by oral, parenteral, subcutaneous, intravenous, intramuscular, intranasal, intrapulmonary, intraarterial, intrathecal, or interperitoneal administration.

(33) The method of any one of aspects 30-32, wherein the coronavirus is selected from the group consisting of SARS-CoV-2, SARS-CoV, and MERS-CoV.

(34) The method of any one of aspects 30-33, wherein the subject is a mammal, preferably a human or non-human primate.

(35) The use of the recombinant polypeptide of any one of aspects 1-18, the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, or the pharmaceutical composition of any one of aspects 23-27 for the preparation of a medicament for the treatment or prevention of illness caused by SARS-CoV-2.

(36) The recombinant polypeptide of any one of aspects 1-18, the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, or the pharmaceutical composition of any one of aspects 23-27 for use as a medicament.

EXAMPLES

The following descriptions of cloning and protein expression apply to each example.

Cloning: Inserts encoding a certain fragment of SARS CoV-2 virus Spike protein 51 were designed based prediction and previous Lytic Solutions expression data. Codon optimized cDNA was synthesized. The insert was cloned using either (a) restriction digests and DNA ligations or (b) NEB HiFi DNA assembly builder mix into a pcDNA3.1 vector containing appropriate secretion signal sequences, linkers, and tags for secreted fusion protein expression, as well as sequence encoding a human IgG1 Fc region. The resulting clone encoding a recombinant SARS CoV-2 Spike protein fragment-Fc region fusion protein ("recombinant CoV-2 fusion protein") was verified by either colony PCR and/or restriction digests. DNA sequencing was also used.

Protein Expression: Supercoiled plasmid of the verified clone was transiently transfected into CHO-S cells and expressed under the control of a constitutive promoter within cell culture conditions ranging from 30-37° C. and 3-10 days in CO₂ (8%) incubators with rotary shaking agitation at speeds of 150 RPM. Cells were removed by centrifugation and culture medium containing the recombinant CoV-2 fusion protein were passed over Protein A agarose to bind the Fc region-containing CoV-2 protein. Filtration could also have been used to remove cells. The column containing bound recombinant CoV-2 fusion protein was washed with phosphate buffered saline. The recombinant CoV-2 fusion protein bound to the Protein A column was eluted with low pH glycine followed by neutralization in pH8.0 Tris. The recombinant CoV-2 fusion protein was dialyzed to remove glycine/tris and placed into HEPES-buffered saline (10 mM HEPES, 150 mM NaCl, pH adjusted with NaOH to pH 7.2-7.5). No additional purification was employed in this case. However, additional purification by any chromatography method such as HIC or ion exchange can optionally be used.

Example 1

SARS CoV-2 antigens were selected from the genomic sequence (ncbi.nlm.nih.gov/nuccore/MN908947) to use in generating antibody and T-cell responses to the receptor binding domains of SARS CoV-2 virus Spike protein. The selected domain encoding regions were codon optimized for CHO cell expression using IDTDNA codon optimization algorithms. Template DNA was synthesized at Twist Bio. A modified pcDNA3.1 vector was used for protein expression that included human IgG1 Fc for translational fusion generation, i.e., a fusion protein containing a SARS CoV-2 virus Spike protein fragment and an antibody Fc region. The vector encoding the fusion protein was expressed in CHO cells, in which the fusion protein was secreted to the media, cells and cell debris were removed by centrifugation, the recombinant protein was captured with Protein A resin, and eluted with low pH glycine buffer. The resulting fusion protein was buffer exchanged by dialysis and mixed with adjuvant. The resulting vaccine composition was injected intramuscularly into Cynomolgus monkeys. The vaccine composition generated an unexpectedly strong immune response in Cynomolgus monkeys.

Vaccination: Following protein expression in accordance with the description above, the purified recombinant CoV-2 fusion protein (LS2330, CTD_short-a-Fc; SEQ ID NO: 73) was mixed with Titermax Gold adjuvant, a modified squalene in water emulsion adjuvant, according to manufacturer directions, and injected intramuscularly into the thigh of Cynomolgus monkeys. Injections were performed at day 0 and day 14. Dosages were 250 ug of antigen of CTD_short-a-Fc (Seq ID NO:73). Prior to the first injection on day 0, a baseline sample was collected from each test subject.

The immune response was monitored every two weeks following the injection on day 0, with the earliest sample taken on day 14. Accordingly, samples were collected on day 14 and day 28. The samples were analyzed, and the results are shown in FIG. 1. Venous blood was obtained from Cynomolgus monkeys pre immunization in EDTA-containing vacutubes (value shown as Day 0 serum sample in FIG. 1). Immediately after the pre-immunization blood sample was taken, monkeys were immunized with CTD_short-a-Fc (Seq ID NO:73). Serum was isolated by centrifugation of the non-coagulated blood. Serum was diluted 1:100 and analyzed on Intuitive Biosciences ELISA platform for anti-S1 spike binding antibodies. Additional time points were taken at 2 weeks post Day 0, 4 weeks post Day 0, and 6 weeks post Day 0. The single immune boost (CTD_short-a-Fc (Seq ID NO:73)) was performed immediately after the 2 week blood sample was taken. Samples were processed similar to Day 0 samples and immune response was recorded as Relative Intensity Units (RIU) on Intuitive Bioscience ELISA platform. Capture of anti-SARS CoV-2 antibodies was performed by spotting SARS CoV-2 S1 spike protein (Sino Biologicals) onto the wells of Intuitive Bioscience ELISA platform 96 well plates, adding diluted serum—full concentration serum and dilutions to 1:200 serum:buffer were used, but higher concentrations of serum surprisingly resulted in a signal too strong to read on the platform, thus requiring a 1:200 fold dilution for the readings shown in FIG. 1—and incubating for 10-120 minutes to allow binding of anti-CTD antibodies to the S1 spike protein attached to the plate wells. Serum was washed 3× from the wells, and followed by application of anti-cynomolgus detection antibody to the well, incubated for 10-120 minutes, then washed 3× from

the well. Detection reagents were added and the signal was quantified compared to control spots of unrelated proteins.

FIG. 1 depicts an unexpectedly strong immune response in Cynomolgus monkeys to the tested construct, particularly in comparison to the immune response generated by other known SARS-CoV-2 constructs. See, for instance, Graham et al., "Evaluation of the immunogenicity of prime-boost vaccination with the replication-deficient viral vectored COVID-19 vaccine candidate ChAdOx1 nCoV-19," bioRxiv preprint doi: doi.org/10.1101/2020.06.20.159715 (posted Jun. 20, 2020)

Neutralization Assays: FIGS. 5A-C depict the results of neutralization assays done on the tested Cynomolgus monkeys P0101 and P0102 (respectively depicted as "Animal 101" and "Animal 102" in FIG. 1). Regarding FIGS. 5A and 5B, "wpi" in the legend denotes weeks post injection/immunization. SARS-CoV-2 pseudotyped particles were generated as previously described (see Schmidt, F., et al. Measuring SARS-CoV-2 neutralizing antibody activity using pseudotyped and chimeric viruses. *J Exp Med*, v. 217, n. 11, 11 2020). Briefly, 293T cells were transfected with pHIV-1NLGagPol, pCCNG/nLuc and pSARS-CoV-2-SA19. Particles were harvested 48 hours after transfection, filtered and stored at -80°C . Fourfold serially diluted serum from the immunized monkeys was incubated with SARS-CoV-2 pseudotyped virus for 1 h at 37°C . The mixture was subsequently incubated with 293T/ACE2c1.22 cells (plated on Poly-D-Lysine-coated 96-well plates) with the final starting dilution of serum being 1:50. At 48 h later the cells were washed with PBS and lysed with Luciferase Cell Culture Lysis 5 $\times$ reagent (Promega). Nanoluc Luciferase activity in lysates was measured using the Nano-Glo Luciferase Assay System (Promega) with the Modulus II Microplate Reader (Turner BioSystems). The raw nanoluc luciferase activity values (relative luminescence units) were normalized to those derived from cells infected with SARS-CoV-2 pseudotyped virus in the absence of serum or a rabbit monoclonal antibody diluted in normal human serum at 0.105 mg/mL (40592-R001, Sinobiological, Wayne, PA). The half-maximal inhibitory concentration for serum (NT50) was determined using four-parameter nonlinear regression (GraphPad Prism).

Immunization of the two Cynomolgus macaques (IDs: P0101 and P0102) with SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]) produced robust neutralizing antibody response. Neutralization assays were performed using a replication-defective single-cycle pseudotyped virus carrying SARS-CoV-2 spikes and the NanoLuc luciferase reporter. This assay has been previously shown to accurately

predict serum neutralizing activity against authentic SARS-CoV-2 (see Schmidt, F., et al. referenced above). As a control for neutralization sensitivity, human serum obtained from a SARS-CoV-2 negative individual was used alone or spiked with a monoclonal neutralizing antibody (FIG. 5C). Serum samples collected at the various timepoints from 0 to 20 weeks post-immunization were evaluated for neutralizing activity. Sera from animals immunized with SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]) had readily detectable neutralization activity, as early as 2 weeks post-immunization that significantly increased until weeks 4 to 8 of the study. Indeed, neutralizing titers were exceptionally high at 4-8 weeks after immunization, in the range of 10,000 to 100,000 and were maintained in the 1000 to 10,000 range at 20 weeks after immunization.

Example 2

Overview: SARS CoV-2 antigens were selected from the SARS CoV-2 genomic sequence (ncbi.nlm.nih.gov/nuccore/MN908947) to use in generating antibody and T-cell responses to the receptor binding domains of SARS CoV-2 virus Spike protein. Preliminary data identified select individual regions of the SARS CoV-2 virus Spike protein that would be amenable to high-level expression as Fc-fusions, that are resistant to proteolysis when expressed in CHO cells, and that generate strong immune response in Cynomolgus macaques. We further determined that in-series multimerization of select SARS CoV-2 virus Spike protein domains can be used to create proteins that retain high-level expression without significant proteolysis sensitivity whilst doubling the theoretical antigenicity of the molecule to be used for immune stimulation. The design of the multiple domain molecules provides a scaffold for straight-forward modification to incorporate amino acid mutations identified in new and emerging variants of the SARS CoV-2 virus Spike protein. The resulting vaccine composition of the Wuhan variant was injected intramuscularly into Rhesus macaques and elicit stronger immune responses using simple adjuvants at doses at mere fractions of that needed with single domain Fc-fusions.

Following protein expression in accordance with the description above, resulting protein was analyzed by reducing SDS-PAGE to determine protein integrity for monomeric CTD-Fcs (FIGS. 2A-D), in-series dimer CTD-Fcs (FIG. 3) and in-series dimer mutant CTD-Fcs (FIG. 4). Protein yields were calculated using absorbance of 280 nm light and protein-specific extinction coefficients determined in silico according to their expected amino acid composition. Representative protein yields are given in Table 1.

TABLE 1

Strain(s)	Insert	Expression (mg protein/L CHO culture ($\pm$ SD))
3472 (SEQ ID NO: 75)	CTD_long_a (SEQ ID NO: 1)	77.3 ⁽⁵⁾
2356 (SEQ ID NO: 101)	CTD_long_b (SEQ ID NO: 7)	60.8 ⁽⁵⁾
2357 (SEQ ID NO: 103)	CTD_long_c (SEQ ID NO: 9)	60.8 ⁽⁵⁾
2358 (SEQ ID NO: 105)	CTD_long_d (SEQ ID NO: 11)	65.8 ⁽⁵⁾
2359 (SEQ ID NO: 107)	CTD_long_e (SEQ ID NO: 13)	96.1 ⁽⁵⁾
2360 (SEQ ID NO: 109)	CTD_long_f (SEQ ID NO: 15)	95.1 ⁽⁵⁾
2361 (SEQ ID NO: 111)	CTD_long_g (SEQ ID NO: 17)	74.5 ⁽⁵⁾
2362 (SEQ ID NO: 113)	CTD_long_h (SEQ ID NO: 19)	108.3 ⁽⁵⁾
2363 (SEQ ID NO: 115)	CTD_long_i (SEQ ID NO: 21)	108.5 ⁽⁵⁾
2330 (SEQ ID NO: 73)	CTD_short_a (SEQ ID NO: 23)	>40 ⁽⁵⁾
2364 (SEQ ID NO: 117)	CTD_short_b (SEQ ID NO: 25)	131.8 ⁽⁵⁾
2365 (SEQ ID NO: 119)	CTD_short_c (SEQ ID NO: 27)	126.5 ⁽⁵⁾
2366 (SEQ ID NO: 121)	CTD_short_d (SEQ ID NO: 29)	106.2 ⁽⁵⁾
2367 (SEQ ID NO: 123)	CTD_short_e (SEQ ID NO: 31)	107.5 ⁽⁵⁾
2368 (SEQ ID NO: 125)	CTD_short_f (SEQ ID NO: 33)	109.9 ⁽⁵⁾

TABLE 1-continued

Strain(s)	Insert	Expression (mg protein/L CHO culture ($\pm$ SD))
2369 (SEQ ID NO: 127)	CTD_short_g (SEQ ID NO: 35)	132.4 ⁽⁵⁾
2370 (SEQ ID NO: 129)	CTD_short_h (SEQ ID NO: 37)	67.5 ⁽⁵⁾
2371 (SEQ ID NO: 131)	CTD_short_i (SEQ ID NO: 39)	167.7 ⁽⁵⁾
2372 (SEQ ID NO: 133)	CTD_vs_a (SEQ ID NO: 41)	2.3 ⁽⁵⁾
2373 (SEQ ID NO: 135)	CTD_vs_b (SEQ ID NO: 43)	1.7 ⁽⁵⁾
2374 (SEQ ID NO: 137)	CTD_vs_c (SEQ ID NO: 45)	0.9 ⁽⁵⁾
2375 (SEQ ID NO: 139)	CTD_vs_d (SEQ ID NO: 47)	≤ 0.8 ⁽⁵⁾
2376 (SEQ ID NO: 141)	CTD_vs_e (SEQ ID NO: 49)	≤ 0.8 ⁽⁵⁾
2377 (SEQ ID NO: 143)	RBD_a (SEQ ID NO: 51)	≤ 0.8 ⁽⁵⁾
2378 (SEQ ID NO: 145)	RBD_b (SEQ ID NO: 53)	≤ 0.8 ⁽⁵⁾
2380 (SEQ ID NO: 149)	RBD_d (SEQ ID NO: 55)	≤ 0.8 ⁽⁵⁾
2381 (SEQ ID NO: 151)	RBD_e (SEQ ID NO: 57)	14.5 ⁽⁵⁾
2393 (SEQ ID NO: 155)	(RBD-tight) ₂ (SEQ ID NO: 173)	≤ 0.8 ⁽⁷⁾
2394 (SEQ ID NO: 157)	(RBD-extended) ₂ (SEQ ID NO: 177)	7.4 ⁽⁷⁾
2395 (SEQ ID NO: 159)	(RBD) ₂ (SEQ ID NO: 181)	≤ 0.8 ⁽⁷⁾
2397-2400 (SEQ ID NO: 161)	(CTD_short_d) ₂ (SEQ ID NO: 183)	32.2 ⁽⁴⁾ /53.0 (± 4) ⁽⁷⁾
2401-2404 (SEQ ID NO: 163)	(CTD_short_i) ₂ (SEQ ID NO: 185)	88.3 ⁽⁴⁾ /144.7 (± 5.5) ⁽⁷⁾
2421 (SEQ ID NO: 165)	[(CTD_short_i) ₂ -Fc; D2 mutations for 501.V2 variant, K417N, E484K, N501Y] (SEQ ID NO: 187)	79.5 ⁽⁴⁾
2423 (SEQ ID NO: 167)	[(CTD_short_i) ₂ -Fc; D2 mutations for hybrid P.1 and CAL.20C variants; K417T, L452R, E484K, N501Y] (SEQ ID NO: 189)	55.2 ⁽⁴⁾
2435 (SEQ ID NO: 169)	[(CTD_short_i) ₂ -Fc; D1 and D2 mutations for hybrid P.1 and CAL.20C variants; K417T, L452R, E484K, N501Y] (SEQ ID NO: 191)	61.2 ⁽⁴⁾

⁽⁴⁾ 4-day expression,⁽⁵⁾ 5-day expression,⁽⁷⁾ 7-day expression

Based on the desire to develop methods to increase the relative antigen content of a protein molecule, in-series concatemers of RBD-containing protein fragments translationally fused to an Fc. CTD_short_d and CTD_short_i (SEQ ID NOs: 29 and 39, respectively) were prepared and selected for in-series expression given their high-level, protease resistant expression as monomers. When conjoined with a serine-glycine linker and expressed in CHO cells, the resulting double-domain proteins expressed at similar levels as the monomer constructs. In addition, the in-series design retained the molecular resistance to proteolysis during expression and purification. Therefore, by combining intramolecular dimerization driven by Fc interactions with in-series domain expression, it was possible to go from a single SARS-CoV-2 antigen fragment per molecule to having 4 (and potentially more) per molecule. The higher antigen content in conjunction with robust expression levels and protein stability provides a suitable framework for an immunogen to be used for vaccine and boost applications.

As an example of this approach, double-domain Fc-fusion constructs (i.e., recombinant polypeptides containing two immunogenic fragments connected to an antibody Fc region) containing amino acid mutations within one or both of the domains were prepared. The mutations used were either from a single virus variant or a hybrid of two virus variants. Incorporation of the mutations into the double-domain-Fc wild-type molecule (strain 2401) depicted in FIG. 3 followed by expression in CHO cells resulted in intact, soluble protein with similar yields to the wild-type amino acid sequence. These results support the concept that this multi-domain Fc-fusion platform provides a robust scaffold for the incorporation and expression molecules that reflect new variant mutations. This provides a robust, timely and cost-effective system to adapt vaccine composition to meet the needs to mitigate evolving variants. A vaccine containing strain 2401 was subsequently tested in Rhesus macaque monkeys.

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Vaccination: Vaccination and boosters to nCoV-2 double dimer (Lytic Solutions Strain #2401, SEQ ID NO: 163, administered to all animals except for those indicated to be administered the nCoV-2 quadruple mutant vaccine) or nCoV-2 quadruple mutant vaccine (Lytic Solutions Strain #2435, SEQ ID NO: 169, referred to in Table 2C as "Variant COVID Vaccine" in Adjuvant & Dose column) were performed on Rhesus macaque monkeys by the following methodology. 50, 25, or 12.5 micrograms (these numbers are used to define dosage) of nCoV-2 protein was mixed with either AS03 (Invivogen catalog vac-as03-10) or alum (Invivogen catalog vac-alu-250) as a 1:1 volume mixture protein: adjuvant. For 50 ug doses 500 ul of each protein and adjuvant were used, for 25 ug and 12.5 ug 250 ul of each protein and adjuvant were used. Dosages were split and injected intramuscularly into left and right thighs at the time of initial vaccination and at booster vaccination. Initial vaccination day was designated day 0. Boosters were given 28 day post day 0 unless otherwise noted. Serum samples were taken prior to initial vaccination on day 0, day 14, day 28 prior to booster vaccination, and day 42 and day 56 were 2/4 weeks post booster injection. Therefore data from 14 and 28 (immunization plus 2 weeks and immunization plus 4 weeks, respectively) days are specific to a single dose of vaccine whereas samples from day 42 and day 56 are 2 doses of vaccine (booster plus 2 weeks, and booster plus 4 weeks respectively). Each animal received the same dosage of protein and adjuvant for the booster dose that they received in the primary dose.

Analysis of antibody titers were performed at Intuitive Biosciences, Madison WI on their proprietary ELISA system (918 Deming Way, Suite 100, Madison WI 53719 USA). Serum was serially diluted in CSA buffer (Intuitive product no. 7-1037) to dilutions of 1:100, 1:1000, 1:10,000, 1:100,000, and 1:1,000,000. Analysis was performed at and by Intuitive Biosciences. ELISA units are measured as density on their platform with a maximum signal of approximately

45,000-50,000 counts. Titer signals were determined from dilutions that yielded signals less than 1/3rd maximal signal. Fifteen animals were used to determine vaccine/booster efficacy. Animal names are codes generated for each animal at the primate facility at UW-Madison. All animals were assayed through day 56 post initial vaccination. 50, 25, and 12.5 ug doses gave similar titers for day 28. Some animals were followed past the study design point of 56 days to various time points up to 23 weeks post initial vaccine (due to continued potency of vaccine response). Data is summarized in Tables 2A, 2B, and 2C. It is noted that, although some of the Plate Sample ID numbers are overlapping, e.g., Tables 2A and 2B both have a row with a plate sample ID of 59, this is merely an artifact of the data collection process, such that rows with the same Plate Sample ID represent independently collected data points.

50 ug, 25 ug, 12.5 ug: These data show that relatively low dosages of the tested constructs, including 12.5 ug, still elicited sufficient immune stimulation and/or boosting. Such low amounts of protein per vaccine dose allow for an increased number of active doses to be produced per liter of cell culture. As production levels increase, not only do cost of goods go down, but the timeframe to produce large numbers of doses can be decreased in comparison to vaccine compositions requiring higher amounts of protein per dose to provide sufficient immune protection. This is particularly relevant to providing immune protection against any newly-arising SARS-CoV-2 strains.

The assembly of four point-mutations into a single RBD polypeptide was undertaken prior to public disclosure of the

Delta mutant isolated first in India. Vaccine LS2435 (SEQ ID NO: 169) contains mutations in four sites that reflect mutation that evolved from new variants identified from the UK, South Africa, Brazil and southern California. Convergent evolution of mutations in new variants lead this to be an attractive approach of stacking multiple mutations in one construct to represent multiple variants, such as seen in the Delta variant. Each RBD point mutation was determined to add virulence through either immune system avoidance or enhanced viral entry or production of higher viral loads (or a combination thereof). The identified mutations were combined to develop a vaccine that could address each mutation and immune epitope singly or in combination. When evolutionary boundaries are considered, it became clear that mutational stacking was likely to take place through either recombination of previous viral strains, or additional mutations stacked onto previous viral strains that enhanced virulence/transmission. Considering the mutation-stacked SARS CoV-2 strains can evade the immune response as well as generate a more potent viral titer, having a vaccine that displays high levels of antibody and T-cell potency is a major advantage over previous COVID-19 vaccines which can provide less potent immune responses to these mutants. The SARS CoV-2 mutant-containing vaccine LS2435 demonstrates highly potent immune stimulation and antibody production in Rhesus macaques. In fact, the levels of antibodies achieved with the mutant vaccine are similar to responses seen from the tested wild-type vaccine, i.e., LS2401.

TABLE 2A

Plate Sample ID	Spike S1	Spike S2	Nucleocapsid	Dilution	Animal - Adjuvant & Dosage	Days post injection
59	14	217	296	1:100	BH56 - AS03 50 ug	0
60	19	21	418	1:1,000	BH56 - AS03 50 ug	0
61	0	30	100	1:10,000	BH56 - AS03 50 ug	0
62	44982	343	489	1:100	BH56 - AS03 50 ug	14
63	31862	49	445	1:1,000	BH56 - AS03 50 ug	14
64	3227	28	276	1:10,000	BH56 - AS03 50 ug	14
65	47053	507	642	1:100	BH56 - AS03 50 ug	28
66	35647	42	46	1:1,000	BH56 - AS03 50 ug	28
67	9818	57	290	1:10,000	BH56 - AS03 50 ug	28
68	34	116	508	1:100	BC43 - Alum 50 ug	0
69	0	30	291	1:1,000	BC43 - Alum 50 ug	0
70	22	52	1	1:10,000	BC43 - Alum 50 ug	0
71	36334	67	208	1:100	BC43 - Alum 50 ug	14
72	14375	720	553	1:1,000	BC43 - Alum 50 ug	14
73	234	0	0	1:10,000	BC43 - Alum 50 ug	14
74	46731	20	36	1:100	BC43 - Alum 50 ug	28
75	39535	56	258	1:1,000	BC43 - Alum 50 ug	28
76	5458	17	1	1:10,000	BC43 - Alum 50 ug	28
77	0	97	318	1:100	BH95 - Alum 50 ug	0

TABLE 2A-continued

Plate Sample ID	Spike S1	Spike S2	Nucleocapsid	Dilution	Animal - Adjuvant & Dosage	Days post injection
78	0	66	146	1:1,000	BH95 - Alum 50 ug	0
79	1	12	23	1:10,000	BH95 - Alum 50 ug	0
80	42119	63	259	1:100	BH95 - Alum 50 ug	14
81	30356	59	644	1:1,000	BH95 - Alum 50 ug	14
82	451	0	0	1:10,000	BH95 - Alum 50 ug	14
83	49046	156	63	1:100	BH95 - Alum 50 ug	28
84	40830	0	187	1:1,000	BH95 - Alum 50 ug	28
85	7282	28	78	1:10,000	BH95 - Alum 50 ug	28
86	40	55	557	1:100	BG86 - AS03 50 ug	0
87	40	93	766	1:1,000	BG86 - AS03 50 ug	0
88	0	27	229	1:10,000	BG86 - AS03 50 ug	0
89	27694	144	618	1:100	BG86 - AS03 50 ug	14
90	436	114	275	1:1,000	BG86 - AS03 50 ug	14
91	243	23	91	1:10,000	BG86 - AS03 50 ug	14
92	46656	85	618	1:100	BG86 - AS03 50 ug	28
93	38813	84	527	1:1,000	BG86 - AS03 50 ug	28
94	13019	61	65	1:10,000	BG86 - AS03 50 ug	28
Neg Cntl	46	475	986			
Pos Cntl	44473	39802	31871			

TABLE 2B

Plate Sample ID	Spike S1	Spike S2	Nucleocapsid	Dilution	Animal - Adjuvant & Dosage	Time after initial vaccination
10	47883	0	96	1:10,000	BC43 - Alum 50 ug	Day 42
11	25917	0	0	1:100,000	BC43 - Alum 50 ug	Day 42
12	3597	14	0	1:1,000,000	BC43 - Alum 50 ug	Day 42
13	45232	0	60	1:10,000	BC43 - Alum 50 ug	Day 56
14	16743	0	0	1:100,000	BC43 - Alum 50 ug	Day 56
15	2099	0	0	1:1,000,000	BC43 - Alum 50 ug	Day 56
16	47513	156	7	1:10,000	BH56 - AS03 50 ug	Day 42
17	24243	31	21	1:100,000	BH56 - AS03 50 ug	Day 42
18	3976	11	0	1:1,000,000	BH56 - AS03 50 ug	Day 42
19	45909	0	31	1:10,000	BH56 - AS03 50 ug	Day 56
20	18140	26	5	1:100,000	BH56 - AS03 50 ug	Day 56
21	1184	2	0	1:1,000,000	BH56 - AS03 50 ug	Day 56
22	44095	53	0	1:10,000	BH95 - Alum 50 ug	Day 42
23	20940	26	0	1:100,000	BH95 - Alum 50 ug	Day 42
24	1845	2	28	1:1,000,000	BH95 - Alum 50 ug	Day 42

TABLE 2B-continued

Plate Sample ID	Spike S1	Spike S2	Nucleocapsid	Dilution	Animal - Adjuvant & Dosage	Time after initial vaccination
25	45464	67	177	1:10,000	BH95 - Alum 50 ug	Day 56
26	17409	21	5	1:100,000	BH95 - Alum 50 ug	Day 56
27	1281	22	0	1:1,000,000	BH95 - Alum 50 ug	Day 56
28	48319	71	0	1:10,000	BC11 - AS03 50 ug	Day 42
29	30010	10	0	1:100,000	BC11 - AS03 50 ug	Day 42
30	671	14	0	1:1,000,000	BC11 - AS03 50 ug	Day 42
31	16796	4	0	1:10,000	BG86 - AS03 50 ug	Single dose 8 weeks
32	1550	11	56	1:100,000	BG86 - AS03 50 ug	Single dose 8 weeks
33	164	17	6	1:1,000,000	BG86 - AS03 50 ug	Single dose 8 weeks
34	0	64	15	1:1,000	BI37 - Alum 25 ug	Day 0
35	2	5	0	1:10,000	BI37 - Alum 25 ug	Day 0
36	8	6	25	1:100,000	BI37 - Alum 25 ug	Day 0
37	40318	112	0	1:1,000	BI37 - Alum 25 ug	Day 14
38	4456	35	28	1:10,000	BI37 - Alum 25 ug	Day 14
39	655	35	17	1:100,000	BI37 - Alum 25 ug	Day 14
40	49137	127	31	1:1,000	BI37 - Alum 25 ug	Day 28
41	32611	29	2	1:10,000	BI37 - Alum 25 ug	Day 28
42	5526	34	0	1:100,000	BI37 - Alum 25 ug	Day 28
43	0	56	15	1:1,000	BK78 - AS03 25 ug	Day 0
44	36	46	27	1:10,000	BK78 - AS03 25 ug	Day 0
45	45	34	0	1:100,000	BK78 - AS03 25 ug	Day 0
46	26462	49	87	1:1,000	BK78 - AS03 25 ug	Day 14
47	965	122	29	1:10,000	BK78 - AS03 25 ug	Day 14
48	92	28	37	1:100,000	BK78 - AS03 25 ug	Day 14
49	36563	67	31	1:1,000	BK78 - AS03 25 ug	Day 28
50	6352	34	17	1:10,000	BK78 - AS03 25 ug	Day 28
51	917	34	60	1:100,000	BK78 - AS03 25 ug	Day 28
52	1	0	160	1:1,000	BM52- AS03 12.5 ug	Day 0
53	15	64	60	1:10,000	BM52- AS03 12.5 ug	Day 0
54	12	79	49	1:100,000	BM52- AS03 12.5 ug	Day 0
55	17806	684	315	1:1,000	BM52- AS03 12.5 ug	Day 14
56	380	147	35	1:10,000	BM52- AS03 12.5 ug	Day 14
57	96	32	18	1:100,000	BM52- AS03 12.5 ug	Day 14
58	39459	440	29	1:1,000	BM52- AS03 12.5 ug	Day 28
59	4104	56	12	1:10,000	BM52- AS03 12.5 ug	Day 28
60	467	13	26	1:100,000	BM52- AS03 12.5 ug	Day 28
61	19	51	0	1:1,000	BC31 - Alum 12.5 ug	Day 0
62	0	0	2	1:10,000	BC31 - Alum 12.5 ug	Day 0

TABLE 2B-continued

Plate Sample ID	Spike S1	Spike S2	Nucleocapsid	Dilution	Animal - Adjuvant & Dosage	Time after initial vaccination
63	0	74	30	1:100,000	BC31 - Alum 12.5 ug	Day 0
64	9436	123	40	1:1,000	BC31 - Alum 12.5 ug	Day 14
65	426	27	0	1:10,000	BC31 - Alum 12.5 ug	Day 14
66	8	31	0	1:100,000	BC31 - Alum 12.5 ug	Day 14
67	36234	0	36	1:1,000	BC31 - Alum 12.5 ug	Day 28
68	7220	24	14	1:10,000	BC31 - Alum 12.5 ug	Day 28
69	160	9	8	1:100,000	BC31 - Alum 12.5 ug	Day 28
Neg Cntl	221	1837	90			
Neg Cntl	12	46	10			
Pos Cntl	49197	44161	45329			
Pos Cntl	47017	41224	46778			

TABLE 2C

Plate Sample ID	Spike S1	Spike S2	Nucleocapsid	Dilution	Animal	Days after initial injection	Adjuvant & Dose
1A	15544	333	104	1:10,000	BG86	84	AS03 50 ug - single dose
1B	1751	277	255	1:100,000	BG86	84	AS03 50 ug - single dose
1C	481	276	123	1:1,000,000	BG86	84	AS03 50 ug - single dose
2A	35682	375	0	1:10,000	BH56	84	AS03 50 ug
2B	9450	313	259	1:100,000	BH56	84	AS03 50 ug
2C	1697	329	253	1:1,000,000	BH56	84	AS03 50 ug
3A	39567	327	368	1:10,000	BC43	84	Alum 50 ug
3B	9891	95	124	1:100,000	BC43	84	Alum 50 ug
3C	1824	272	0	1:1,000,000	BC43	84	Alum 50 ug
4A	37479	135	718	1:10,000	BH95	84	Alum 50 ug
4B	8790	161	143	1:100,000	BH95	84	Alum 50 ug
4C	1619	201	132	1:1,000,000	BH95	84	Alum 50 ug
6A	8417	127	454	1:10,000	BC11	14	AS03 50 ug
6B	849	179	71	1:100,000	BC11	14	AS03 50 ug
6C	265	381	273	1:1,000,000	BC11	14	AS03 50 ug
7A	13381	607	664	1:10,000	BC11	28	AS03 50 ug
7B	803	176	81	1:100,000	BC11	28	AS03 50 ug
7C	341	251	275	1:1,000,000	BC11	28	AS03 50 ug
8A	45422	341	285	1:10,000	BC11	56	AS03 50 ug
8B	18249	173	165	1:100,000	BC11	56	AS03 50 ug
8C	3752	213	357	1:1,000,000	BC11	56	AS03 50 ug
9A	42056	341	292	1:10,000	BC11	84	AS03 50 ug
9B	14271	316	163	1:100,000	BC11	84	AS03 50 ug
9C	2393	256	129	1:1,000,000	BC11	84	AS03 50 ug
10A	44288	237	150	1:10,000	BI37	42	Alum 25 ug
10B	19432	189	119	1:100,000	BI37	42	Alum 25 ug
10C	4354	378	231	1:1,000,000	BI37	42	Alum 25 ug
11A	43800	239	35	1:10,000	BI37	56	Alum 25 ug
11B	15355	138	75	1:100,000	BI37	56	Alum 25 ug
11C	2865	337	247	1:1,000,000	BI37	56	Alum 25 ug
12A	38022	305	18	1:10,000	BI37	84	Alum 25 ug
12B	11812	294	86	1:100,000	BI37	84	Alum 25 ug
12C	1948	213	243	1:1,000,000	BI37	84	Alum 25 ug
13A	46998	398	167	1:10,000	BK78	42	AS03 25 ug
13B	24787	267	116	1:100,000	BK78	42	AS03 25 ug
13C	6808	442	0	1:1,000,000	BK78	42	AS03 25 ug
14A	45244	441	0	1:10,000	BK78	56	AS03 25 ug
14B	18294	209	84	1:100,000	BK78	56	AS03 25 ug
14C	3830	484	256	1:1,000,000	BK78	56	AS03 25 ug
15A	41106	478	0	1:10,000	BK78	84	AS03 25 ug
15B	14414	275	83	1:100,000	BK78	84	AS03 25 ug
15C	3328	435	269	1:1,000,000	BK78	84	AS03 25 ug
16A	46111	524	779	1:10,000	BM52	42	AS03 12.5 ug
16B	24586	113	300	1:100,000	BM52	42	AS03 12.5 ug
16C	7160	223	156	1:1,000,000	BM52	42	AS03 12.5 ug

TABLE 2C-continued

Plate Sample ID	Spike S1	Spike S2	Nucleocapsid	Dilution	Animal	Days after initial injection	Adjuvant & Dose
17A	41733	328	527	1:10,000	BM52	56	AS03 12.5 ug
17B	18140	284	157	1:100,000	BM52	56	AS03 12.5 ug
17C	3499	222	380	1:1,000,000	BM52	56	AS03 12.5 ug
18A	36263	995	242	1:10,000	BM52	84	AS03 12.5 ug
18B	11912	525	277	1:100,000	BM52	84	AS03 12.5 ug
18C	1679	190	301	1:1,000,000	BM52	84	AS03 12.5 ug
19A	43970	345	37	1:10,000	BC31	42	Alum 12.5 ug
19B	20155	255	25	1:100,000	BC31	42	Alum 12.5 ug
19C	3849	321	268	1:1,000,000	BC31	42	Alum 12.5 ug
20A	40779	264	0	1:10,000	BC31	56	Alum 12.5 ug
20B	18417	298	196	1:100,000	BC31	56	Alum 12.5 ug
20C	2169	261	24	1:1,000,000	BC31	56	Alum 12.5 ug
21A	32251	89	0	1:10,000	BC31	84	Alum 12.5 ug
21B	9977	314	197	1:100,000	BC31	84	Alum 12.5 ug
21C	1521	384	213	1:1,000,000	BC31	84	Alum 12.5 ug
22A	43	249	129	1:10,000	BM21	0	AS03 12.5 ug
22B	267	215	186	1:100,000	BM21	0	AS03 12.5 ug
22C	225	270	141	1:1,000,000	BM21	0	AS03 12.5 ug
23A	2694	331	96	1:10,000	BM21	14	AS03 12.5 ug
23B	621	202	190	1:100,000	BM21	14	AS03 12.5 ug
23C	297	339	225	1:1,000,000	BM21	14	AS03 12.5 ug
24A	5672	361	75	1:10,000	BM21	28	AS03 12.5 ug
24B	905	39	168	1:100,000	BM21	28	AS03 12.5 ug
24C	91	129	139	1:1,000,000	BM21	28	AS03 12.5 ug
25A	45613	555	118	1:10,000	BM21	49	AS03 12.5 ug
25B	23568	513	244	1:100,000	BM21	49	AS03 12.5 ug
25C	4134	293	248	1:1,000,000	BM21	49	AS03 12.5 ug
26A	39959	429	19	1:10,000	BM21	63	AS03 12.5 ug
26B	15663	309	161	1:100,000	BM21	63	AS03 12.5 ug
26C	2263	240	273	1:1,000,000	BM21	63	AS03 12.5 ug
27A	34745	371	47	1:10,000	BM21	91	AS03 12.5 ug
27B	9926	402	121	1:100,000	BM21	91	AS03 12.5 ug
27C	1197	299	380	1:1,000,000	BM21	91	AS03 12.5 ug
28A	879	302	80	1:10,000	BD82	0	Alum 12.5 ug
28B	538	470	120	1:100,000	BD82	0	Alum 12.5 ug
28C	591	135	129	1:1,000,000	BD82	0	Alum 12.5 ug
29A	3764	90	0	1:10,000	BD82	14	Alum 12.5 ug
29B	532	397	227	1:100,000	BD82	14	Alum 12.5 ug
29C	307	583	267	1:1,000,000	BD82	14	Alum 12.5 ug
30A	9816	337	84	1:10,000	BD82	21	Alum 12.5 ug
30B	2407	371	169	1:100,000	BD82	21	Alum 12.5 ug
30C	285	462	245	1:1,000,000	BD82	21	Alum 12.5 ug
31A	41086	344	128	1:10,000	BD82	35	Alum 12.5 ug
31B	19882	159	219	1:100,000	BD82	35	Alum 12.5 ug
31C	2912	333	73	1:1,000,000	BD82	35	Alum 12.5 ug
32A	36345	245	57	1:10,000	BD82	49	Alum 12.5 ug
32B	13460	469	264	1:100,000	BD82	49	Alum 12.5 ug
32C	2585	149	9	1:1,000,000	BD82	49	Alum 12.5 ug
33A	27977	400	185	1:10,000	BD82	77	Alum 12.5 ug
33B	7948	288	80	1:100,000	BD82	77	Alum 12.5 ug
33C	904	262	28	1:1,000,000	BD82	77	Alum 12.5 ug
34A	62	274	150	1:10,000	BE83	0	Alum 25 ug
34B	385	232	217	1:100,000	BE83	0	Alum 25 ug
34C	142	306	292	1:1,000,000	BE83	0	Alum 25 ug
35A	3232	69	94	1:10,000	BE83	14	Alum 25 ug
35B	804	343	34	1:100,000	BE83	14	Alum 25 ug
35C	129	222	161	1:1,000,000	BE83	14	Alum 25 ug
36A	8300	150	58	1:10,000	BE83	21	Alum 25 ug
36B	1700	295	153	1:100,000	BE83	21	Alum 25 ug
36C	266	285	111	1:1,000,000	BE83	21	Alum 25 ug
37A	38645	368	0	1:10,000	BE83	35	Alum 25 ug
37B	13433	320	151	1:100,000	BE83	35	Alum 25 ug
37C	3155	306	112	1:1,000,000	BE83	35	Alum 25 ug
38A	35954	375	67	1:10,000	BE83	49	Alum 25 ug
38B	14846	190	89	1:100,000	BE83	49	Alum 25 ug
38C	3553	339	246	1:1,000,000	BE83	49	Alum 25 ug
39A	23360	373	0	1:10,000	BE83	77	Alum 25 ug
39B	7836	72	31	1:100,000	BE83	77	Alum 25 ug
40A	12128	194	190	1:10,000	BG86	105	AS03 50 ug - single dose
40B	2292	161	159	1:100,000	BG86	105	AS03 50 ug - single dose
40C	261	187	191	1:1,000,000	BG86	105	AS03 50 ug - single dose
41A	10969	161	0	1:10,000	BG86	139	AS03 50 ug - single dose

TABLE 2C-continued

Plate Sample ID	Spike S1	Spike S2	Nucleocapsid	Dilution	Animal	Days after initial injection	Adjuvant & Dose
41B	1875	219	181	1:100,000	BG86	139	AS03 50 ug - single dose
41C	992	837	705	1:1,000,000	BG86	139	AS03 50 ug - single dose
42A	176	57	120	1:10,000	BG86	MILK 1 month post birth	AS03 50 ug - single dose
42B	567	216	123	1:100,000	BG86	MILK 1 month post birth	AS03 50 ug - single dose
42C	217	283	153	1:1,000,000	BG86	MILK 1 month post birth	AS03 50 ug - single dose
43A	9055	140	149	1:10,000	BP49 Mother is BG86	Birth Serum	no vaccine
43B	1410	121	235	1:100,000	BP49 Mother is BG86	Birth Serum	no vaccine
43C	388	311	152	1:1,000,000	BP49 Mother is BG86	Birth Serum	no vaccine
44A	4525	35	0	1:10,000	BP49 Mother is BG86	4 weeks post birth serum	no vaccine
44B	1259	269	40	1:100,000	BP49 Mother is BG86	4 weeks post birth serum	no vaccine
44C	286	261	223	1:1,000,000	BP49 Mother is BG86	4 weeks post birth serum	no vaccine
45A	23973	304	97	1:10,000	BC43	162	Alum 50 ug
45B	4896	272	161	1:100,000	BC43	162	Alum 50 ug
45C	649	212	208	1:1,000,000	BC43	162	Alum 50 ug
46A	32326	192	36	1:10,000	BK78	142	AS03 25 ug
46B	11039	336	26	1:100,000	BK78	142	AS03 25 ug
46C	1694	418	226	1:1,000,000	BK78	142	AS03 25 ug
47A	21598	206	0	1:10,000	BM72	120	AS03 25 ug
47B	5364	274	237	1:100,000	BM72	120	AS03 25 ug
47C	496	305	222	1:1,000,000	BM72	120	AS03 25 ug
48A	18031	166	168	1:10,000	BH95	162	Alum 50 ug
48B	3372	304	173	1:100,000	BH95	162	Alum 50 ug
48C	562	350	191	1:1,000,000	BH95	162	Alum 50 ug
49A	31501	102	147	1:10,000	BC11	150	AS03 50 ug
49B	8850	327	191	1:100,000	BC11	150	AS03 50 ug
49C	1716	447	257	1:1,000,000	BC11	150	AS03 50 ug
50A	38099	414	277	1:10,000	BM52	142	AS03 12.5 ug
50B	16407	288	305	1:100,000	BM52	142	AS03 12.5 ug
50C	3171	304	136	1:1,000,000	BM52	142	AS03 12.5 ug
51A	358	131	191	1:10,000	BJ55	0	None - Control
51B	443	363	357	1:100,000	BJ55	0	None - Control
51C	138	371	117	1:1,000,000	BJ55	0	None - Control
52A	19609	216	139	1:10,000	BI37	150	Alum 25 ug
52B	4058	261	214	1:100,000	BI37	150	Alum 25 ug
52C	465	148	32	1:1,000,000	BI37	150	Alum 25 ug
53A	31360	392	175	1:10,000	BH56	169	AS03 50 ug
53B	8082	202	146	1:100,000	BH56	169	AS03 50 ug
53C	1013	342	97	1:1,000,000	BH56	169	AS03 50 ug
54A	20996	282	110	1:10,000	BE83	118	Alum 25 ug
54B	4122	195	75	1:100,000	BE83	118	Alum 25 ug
54C	534	319	226	1:1,000,000	BE83	118	Alum 25 ug
55A	433	353	151	1:10,000	BI98	0	None - Control
55B	290	186	114	1:100,000	BI98	0	None - Control
55C	129	297	190	1:1,000,000	BI98	0	None - Control
56A	21633	245	129	1:10,000	BI21	56	Variant COVID Vaccine AS03 25 ug

TABLE 2C-continued

Plate Sample ID	Spike S1	Spike S2	Nucleocapsid	Dilution	Animal	Days after initial injection	Adjuvant & Dose
56B	5705	570	216	1:100,000	BI21	56	Variant COVID Vaccine AS03 25 ug
56C	840	228	155	1:1,000,000	BI21	56	Variant COVID Vaccine AS03 25 ug
57A	10191	173	239	1:10,000	BC31	148	Alum 12.5 ug
57B	1927	340	170	1:100,000	BC31	148	Alum 12.5 ug
57C	217	480	353	1:1,000,000	BC31	148	Alum 12.5 ug
58A	25739	262	179	1:10,000	BK44	56	Variant COVID Vaccine AS03 25 ug
58B	4927	278	91	1:100,000	BK44	56	Variant COVID Vaccine AS03 25 ug
58C	1183	183	113	1:1,000,000	BK44	56	Variant COVID Vaccine AS03 25 ug
59A	20341	177	4	1:10,000	BD82	118	Alum 12.5 ug
59B	3719	264	238	1:100,000	BD82	118	Alum 12.5 ug
59C	410	377	318	1:1,000,000	BD82	118	Alum 12.5 ug
60A	38868	621	413	1:10,000	BM52	148	AS03 12.5 ug
60B	14327	303	250	1:100,000	BM52	148	AS03 12.5 ug
60C	1649	214	132	1:1,000,000	BM52	148	AS03 12.5 ug
61A	26273	155	0	1:10,000	BM21	125	AS03 12.5 ug
61B	5330	236	108	1:100,000	BM21	125	AS03 12.5 ug
61C	861	340	169	1:1,000,000	BM21	125	AS03 12.5 ug
62A	19650	161	0	1:10,000	BI21	84	Variant COVID Vaccine AS03 25 ug
62B	5896	307	69	1:100,000	BI21	84	Variant COVID Vaccine AS03 25 ug
62C	747	244	229	1:1,000,000	BI21	84	Variant COVID Vaccine AS03 25 ug
63A	20018	179	39	1:10,000	BK44	84	Variant COVID Vaccine AS03 25 ug
63B	3860	184	53	1:100,000	BK44	84	A Variant COVID Vaccine S03 25 ug
63C	614	201	54	1:1,000,000	BK44	84	Variant COVID Vaccine AS03 25 ug
66A	0	56	131	1:10,000	BK44	0	Variant COVID Vaccine AS03 25 ug
66B	144	450	345	1:100,000	BK44	0	Variant COVID Vaccine AS03 25 ug
66C	224	411	146	1:1,000,000	BK44	0	Variant COVID Vaccine AS03 25 ug
67A	1739	73	60	1:10,000	BK44	28	Variant COVID Vaccine AS03 25 ug
67B	415	306	202	1:100,000	BK44	28	Variant COVID Vaccine AS03 25 ug

TABLE 2C-continued

Plate Sample ID	Spike S1	Spike S2	Nucleocapsid	Dilution	Animal	Days after initial injection	Adjuvant & Dose
67C	19	229	161	1:1,000,000	BK44	28	Variant COVID Vaccine AS03 25 ug
68A	25898	257	0	1:10,000	BK44	56	Variant COVID Vaccine AS03 25 ug
68B	7243	332	446	1:100,000	BK44	56	Variant COVID Vaccine AS03 25 ug
68C	1135	174	92	1:1,000,000	BK44	56	Variant COVID Vaccine AS03 25 ug
69A	11	215	103	1:10,000	BI21	0	Variant COVID Vaccine AS03 25 ug
69B	95	0	50	1:100,000	BI21	0	Variant COVID Vaccine AS03 25 ug
69C	152	165	102	1:1,000,000	BI21	0	Variant COVID Vaccine AS03 25 ug
70A	2679	232	120	1:10,000	BI21	28	Variant COVID Vaccine AS03 25 ug
70B	429	335	2	1:100,000	BI21	28	Variant COVID Vaccine AS03 25 ug
70C	327	1562	839	1:1,000,000	BI21	28	Variant COVID Vaccine AS03 25 ug
71A	22367	177	31	1:10,000	BI21	56	Variant COVID Vaccine AS03 25 ug
71B	5939	281	116	1:100,000	BI21	56	Variant COVID Vaccine AS03 25 ug
71C	841	299	142	1:1,000,000	BI21	56	Variant COVID Vaccine AS03 25 ug
72A	85	292	16	1:10,000	BM72	0	AS03 25 ug
72B	331	449	281	1:100,000	BM72	0	AS03 25 ug
72C	294	312	139	1:1,000,000	BM72	0	AS03 25 ug
73A	4015	229	162	1:10,000	BM72	28	AS03 25 ug
73B	764	251	262	1:100,000	BM72	28	AS03 25 ug
73C	476	499	165	1:1,000,000	BM72	28	AS03 25 ug
74A	35753	239	39	1:10,000	BM72	56	AS03 25 ug
74B	13297	277	123	1:100,000	BM72	56	AS03 25 ug
74C	2518	396	126	1:1,000,000	BM72	56	AS03 25 ug

Example 3

Further constructs containing certain SARS-CoV-2 immunogenic fragments were prepared and expressed in accordance with the methods disclosed herein. See, e.g., the first three paragraphs of the Example section. The protein expression of the constructs was subsequently measured after four days of expression. See Table 3; FIG. 6. All tested constructs expressed at a concentration of at least about 50 mg protein/L culture.

TABLE 3

Protein yields for Protein A affinity-purified BA.X-monomer-Fc proteins after 4-days of expression.			
Strain Number	Protein	mg protein/L culture	
2634	BA.X-1-monomer-Fc (SEQ ID NO: 255)	85.9	
2635	BA.X-2-monomer-Fc (SEQ ID NO: 256)	58.3	

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TABLE 3-continued

Protein yields for Protein A affinity-purified BA.X-monomer-Fc proteins after 4-days of expression.		
Strain Number	Protein	mg protein/L culture
2636	BA.X-3 -monomer-Fc (SEQ ID NO: 257)	59.5
2637	BA.X-4 -monomer-Fc (SEQ ID NO: 258)	69.3
2638	BA.X-5 -monomer-Fc (SEQ ID NO: 259)	47.9
2639	BA.X-6 -monomer-Fc (SEQ ID NO: 260)	51.4

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TABLE 3-continued

Protein yields for Protein A affinity-purified BA.X-monomer-Fc proteins after 4-days of expression.		
Strain Number	Protein	mg protein/L culture
2640	BA.X-7 -monomer-Fc (SEQ ID NO: 261)	73.4
2641	BA.X-8 -monomer-Fc (SEQ ID NO: 262)	91.0
2642	BA.X-9 -monomer-Fc (SEQ ID NO: 263)	48.2
2643	BA.X-10 -monomer-Fc (SEQ ID NO: 264)	75.7

TABLE 4

Mutations within Monomer (with corresponding SEQ ID NO) In Comparison to Wuhan wild-type variant										
Mutation Position	BA.X-1 (245)	BA.X-2 (246)	BA.X-3 (247)	BA.X-4 (248)	BA.X-5 (249)	BA.X-6 (250)	BA.X-7 (251)	BA.X-8 (252)	BA.X-9 (253)	BA.X-10 (254)
G339	G339D	G339D	G339D	G339D	G339D	G339D	G339D	G339D	G339D	G339D
R346	R346K								R346K	
S371			S371F		S371F		S371F	S371L		
S373		S373P	S373P		S373P		S373P			S373P
S375			S375F		S375F		S375F			
T376	T376A	T376A	T376A	T376A	T376A	T376A	T376A		T376A	T376A
D405	D405N	D405N	D405N	D405N	D405N	D405N	D405N		D405N	D405N
R408	R408S	R408S	R408S	R408S	R408S	R408S	R408S		R408S	R408S
K417	K417N	K417N	K417N	K417N	K417N	K417N	K417N	K417N	K417N	K417N
N440	N440K	N440K	N440K	N440K	N440K	N440K	N440K	N440K	N440K	N440K
G446								G446S	G446S	
L452	L452R			L452R	L452R				L452R	
S477	S477N	S477N	S477N	S477N	S477N	S477N	S477N	S477N	S477N	S477N
T478	T478K	T478K	T478K	T478K	T478K	T478K	T478K	T478K	T478K	T478K
E484	E484A	E484A	E484A	E484K	E484K	E484A	E484A	E484A	E484K	E484A
Q493	Q493R	Q493R	Q493R	Q493R	Q493R		Q493R	Q493R	Q493R	Q493R
G496							G496S	G496S		
Q498	Q498R	Q498R	Q498R	Q498R			Q498R	Q498R	Q498R	Q498R
N501	N501Y	N501Y	N501Y	N501Y	N501Y	N501Y	N501Y	N501Y	N501Y	N501Y
Y505	Y505H	Y505H	Y505H	Y505H			Y505H	Y505H	Y505H	Y505H

Table 4 (above) depicts the amino acid substitutions, and corresponding positions thereof, within immunogenic fragments BA.X-1-BA.X-10 (SEQ ID NOs: 245-254) in comparison to the Wuhan wild-type variant. Table 5 (below) depicts the amino acid substitutions, and corresponding positions thereof, within the listed natural SARS-CoV-2 variants. As revealed by comparison of Tables 4 and 5, the BA.X-1-BA.X-10 immunogenic fragments each contain a combination of mutations not present in any shown natural SARS-CoV-2 strain. The exemplary particular combination of mutations selected for each BA.X-1-BA.X-10 immunogenic fragment provides broad antigenic coverage over existing SARS-CoV-2 Spike protein variants. Without being bound to any theory, it is expected that the broad antigenic coverage should also provide full or partial protection against later-arising SARS-CoV-2 Spike protein variants as well. The constructs containing the BA.X-1-BA.X-10 immunogenic fragments are capable of being expressed at levels suitable for commercial vaccine production. See Table 3, FIG. 6.

TABLE 5

Natural variant mutations in comparison to Wuhan wild-type variant								
Mutation Position	Alpha	Beta	Gamma	Delta	Delta Plus	Mu	Omicron B.1.1.529	Omicron BA.2
G339							G339D	G339D
R346						R346K		
S371							S371L	S371F
S373							S373P	S373P
S375							S375F	S375F
T376								T376A
D405								D405N
R408								R408S
K417		K417N	K417T		K417N		K417N	K417N
N440							N440K	N440K
G446							G446S	
L452				L452R	L452R			
S477							S477N	S477N
T478				T478K	T478K		T478K	T478K
E484		E484K	E484K			E484K	E484A	E484A
Q493							Q493R	Q493R
G496							G496S	
Q498							Q498R	Q498R
N501	N501Y	N501Y	N501Y			N501Y	N501Y	N501Y
Y505							Y505H	Y505H

While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. It will be understood that changes and modifications may be made by those of ordinary skill within the scope of the following claims. In particular, the present invention covers further embodiments with any combination of features from different embodiments described above and below. Additionally, statements made herein characterizing the invention refer to an embodiment of the invention and not necessarily all embodiments.

The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted

as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12409219B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

The invention claimed is:

1. A recombinant polypeptide comprising at least one immunogenic fragment of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein comprising an amino acid sequence with at least 93% sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254.

2. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least two SARS-CoV-2 spike glycoprotein immunogenic fragments.

3. The recombinant polypeptide of claim 2, wherein two immunogenic fragments of the at least two immunogenic fragments comprise the same amino acid sequence.

4. The recombinant polypeptide of claim 2, wherein two immunogenic fragments of the at least two immunogenic fragments comprise a different amino acid sequence.

5. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least one immunogenic fragment comprising an amino acid sequence with at least 93% sequence identity to the amino acid sequence of SEQ ID NO: 245.

6. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least one immunogenic fragment comprising an amino acid sequence with at least 93% sequence identity to the amino acid sequence of SEQ ID NO: 248.

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7. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least one immunogenic fragment comprising an amino acid sequence with at least 93% sequence identity to the amino acid sequence of SEQ ID NO: 252.

8. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least one immunogenic fragment comprising an amino acid sequence with at least 93% sequence identity to the amino acid sequence of SEQ ID NO: 254.

9. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least one immunogenic fragment comprising an amino acid sequence with at least 93% sequence identity to the amino acid sequence of an amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254, wherein the at least one immunogenic fragment further comprises one or more of the following amino acid substitutions: G339D, R346K, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, G446S, L452R, S477N, T478K, E484A, E484K, Q493R, G496S, Q498R, N501Y, and Y505H.

10. The recombinant polypeptide of claim 2, wherein the at least two immunogenic fragments each comprise an amino acid sequence with at least 93% sequence identity to the amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254.

11. The recombinant polypeptide of claim 1, wherein the polypeptide comprises an amino acid sequence with at least 93% sequence identity to the amino acid sequence selected from the group consisting of SEQ ID NOs: 255-264.

12. The recombinant polypeptide of claim 1, wherein the polypeptide comprises an amino acid sequence with at least 93% sequence identity to the amino acid sequence selected from the group consisting of SEQ ID NOs: 275-284.

13. The recombinant polypeptide of claim 2, wherein the at least two immunogenic fragments are connected to each

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other via a linker, and wherein the linker is a polypeptide comprising an amino acid sequence of 1-35 residues, wherein each residue is independently serine or glycine.

14. The recombinant polypeptide of claim 1, wherein the at least one immunogenic fragment of the SARS-CoV-2 spike glycoprotein is connected to an antibody Fc region via a linker, wherein the linker comprises an amino acid sequence selected from the group consisting of SEQ ID NO: 65, SEQ ID NO: 67, SEQ ID NO: 69, SEQ ID NO: 193, SEQ ID NO: 195, and SEQ ID NO: 197.

15. A pharmaceutical composition comprising the recombinant polypeptide of claim 1 and a pharmaceutically acceptable carrier.

16. The pharmaceutical composition of claim 15, wherein the pharmaceutical composition comprises an adjuvant.

17. The pharmaceutical composition of claim 15, wherein the pharmaceutical composition does not comprise an adjuvant.

18. A pharmaceutical composition comprising the recombinant polypeptide of claim 2 and a pharmaceutically acceptable carrier.

19. A pharmaceutical composition comprising a recombinant polypeptide, the recombinant polypeptide comprising at least two immunogenic fragments of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein, and wherein at least one of the at least two immunogenic fragments comprises an amino acid sequence with at least 97% sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254.

20. The pharmaceutical composition of claim 19, wherein at least two of the at least two immunogenic fragments comprises an amino acid sequence with at least 97% sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254.

* * * * *